LONDON INSTITUTE OF SPACE POLICY AND LAW

SPACE POLICY AND LAW COURSE

ON-LINE
9 to 11 November 2020

SPACE HAZARDS AND SAFETY

Professor Richard Crowther
Chief Engineer, UK Space Agency
ISPL Faculty and Adviser

Near Earth Objects

Prof. Richard Crowther, UK



Outline

- Near Earth Objects
  - What are Near Earth Objects?
  - What are the hazards?
  - Is the Earth at risk?
  - What are the issues?

What are Near Earth Objects?



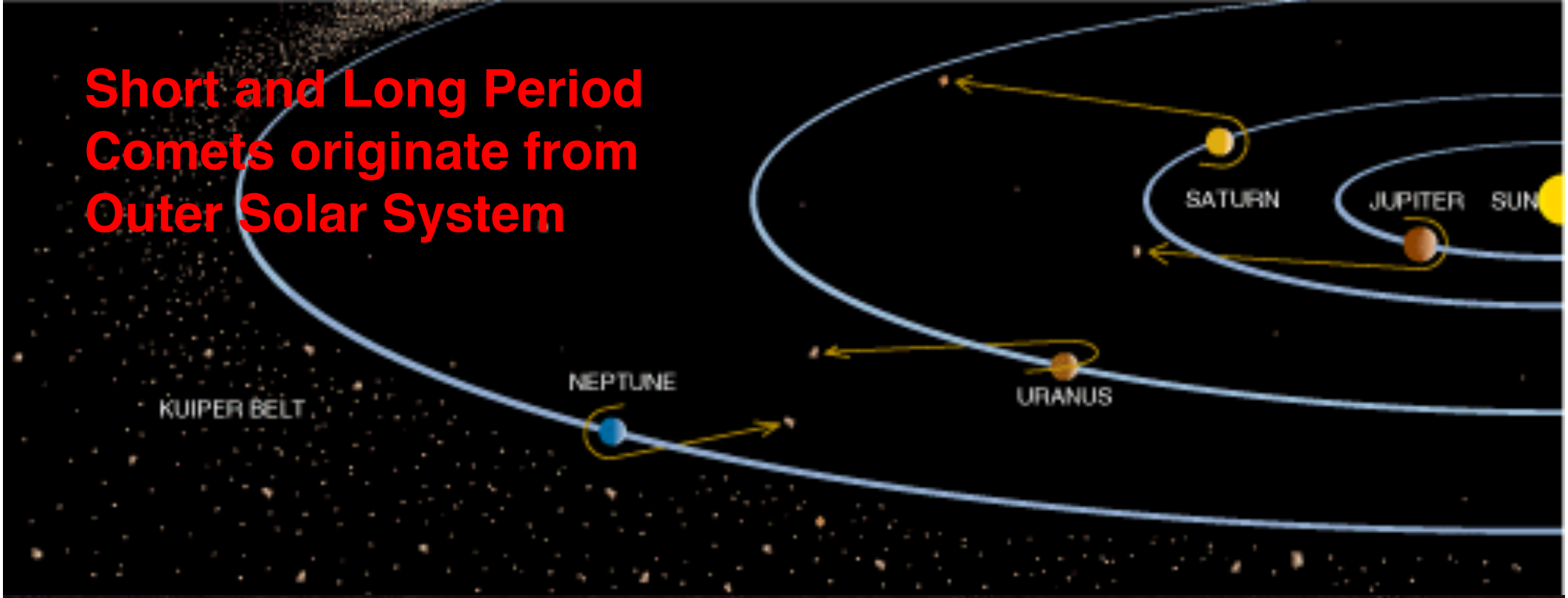


COMET or ASTEROID

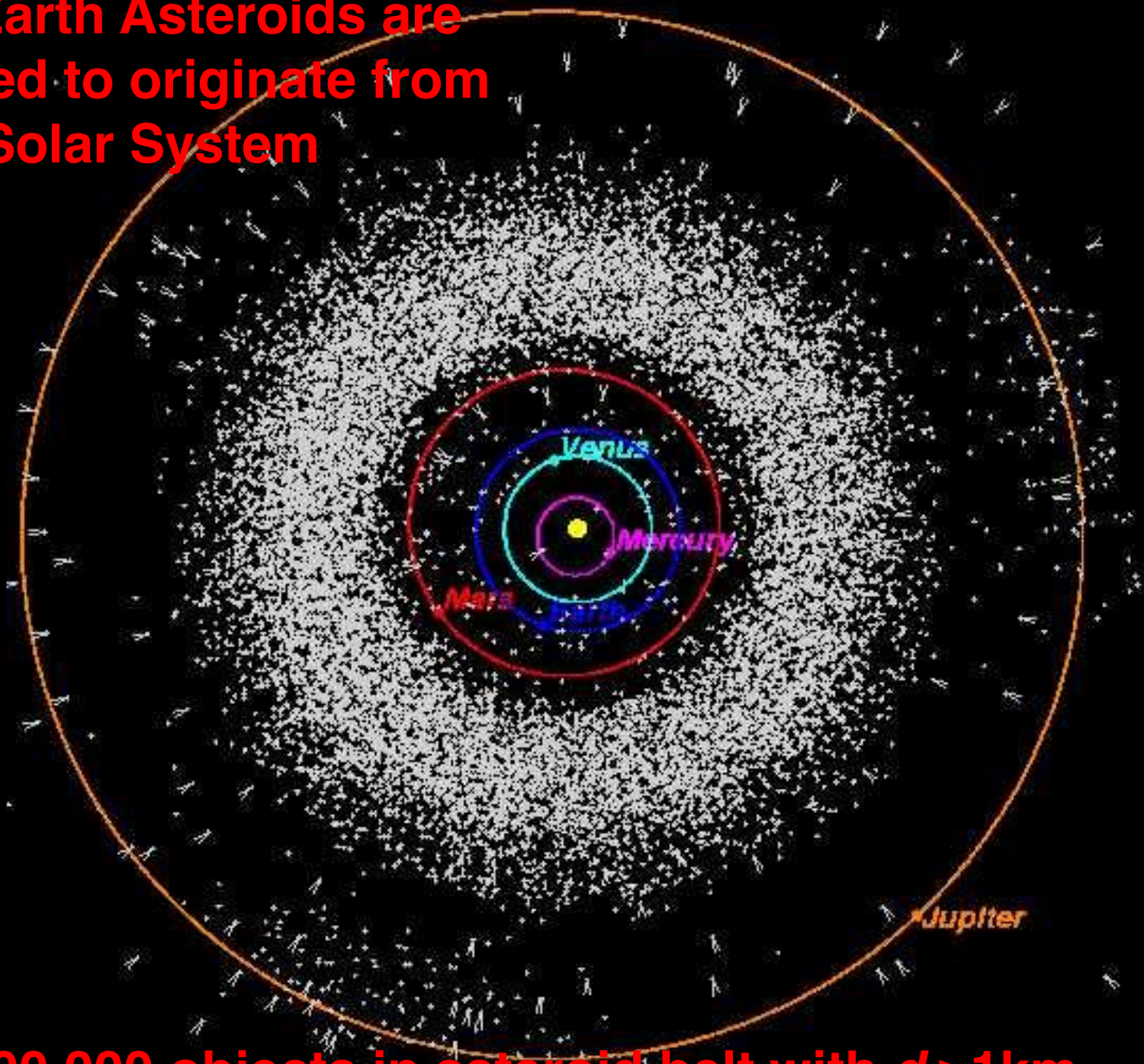
with orbit within 0.3 AU of Earth

1 AU = 150,000,000 km

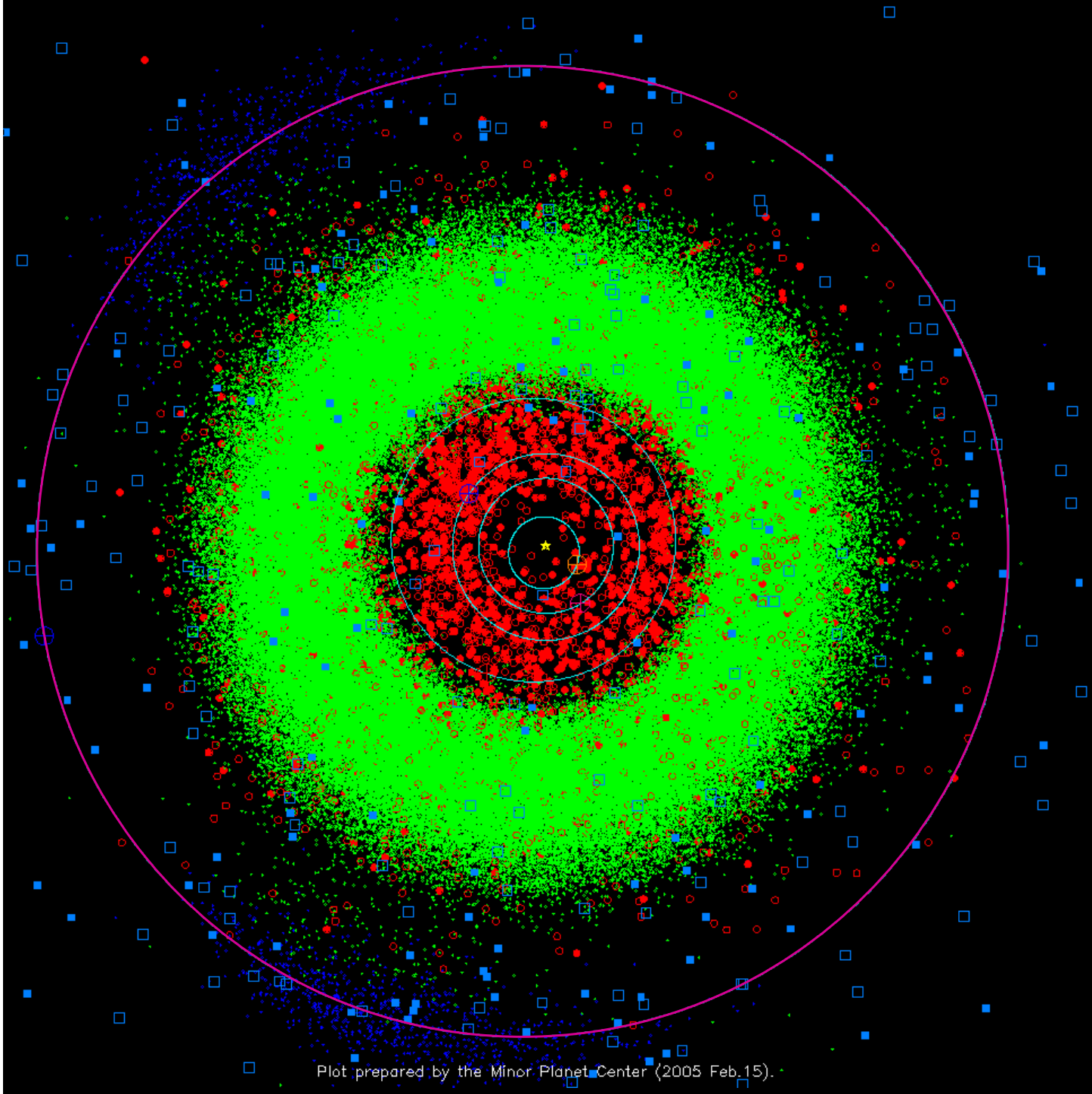
Potentially Hazardous if within 0.05 AU (7,500,000 km) and size > 150m


Short and Long Period
Comets originate from
Outer Solar System
SATURN
JUPITER
SUN
NEPTUNE
URANUS
KUIPER BELT

Near Earth Asteroids are believed to originate from Inner Solar System



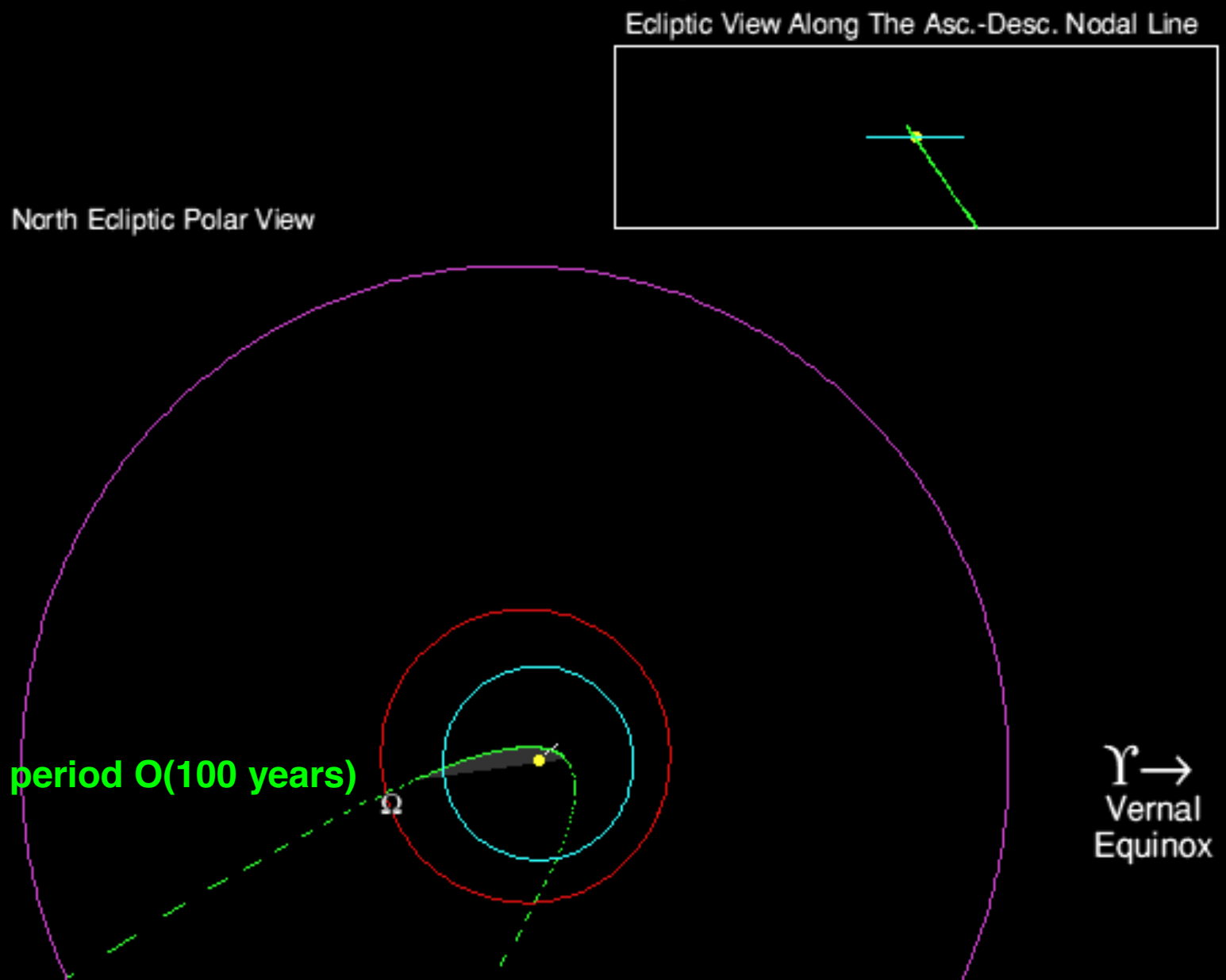
1,000,000 objects in asteroid belt with *d* >1km



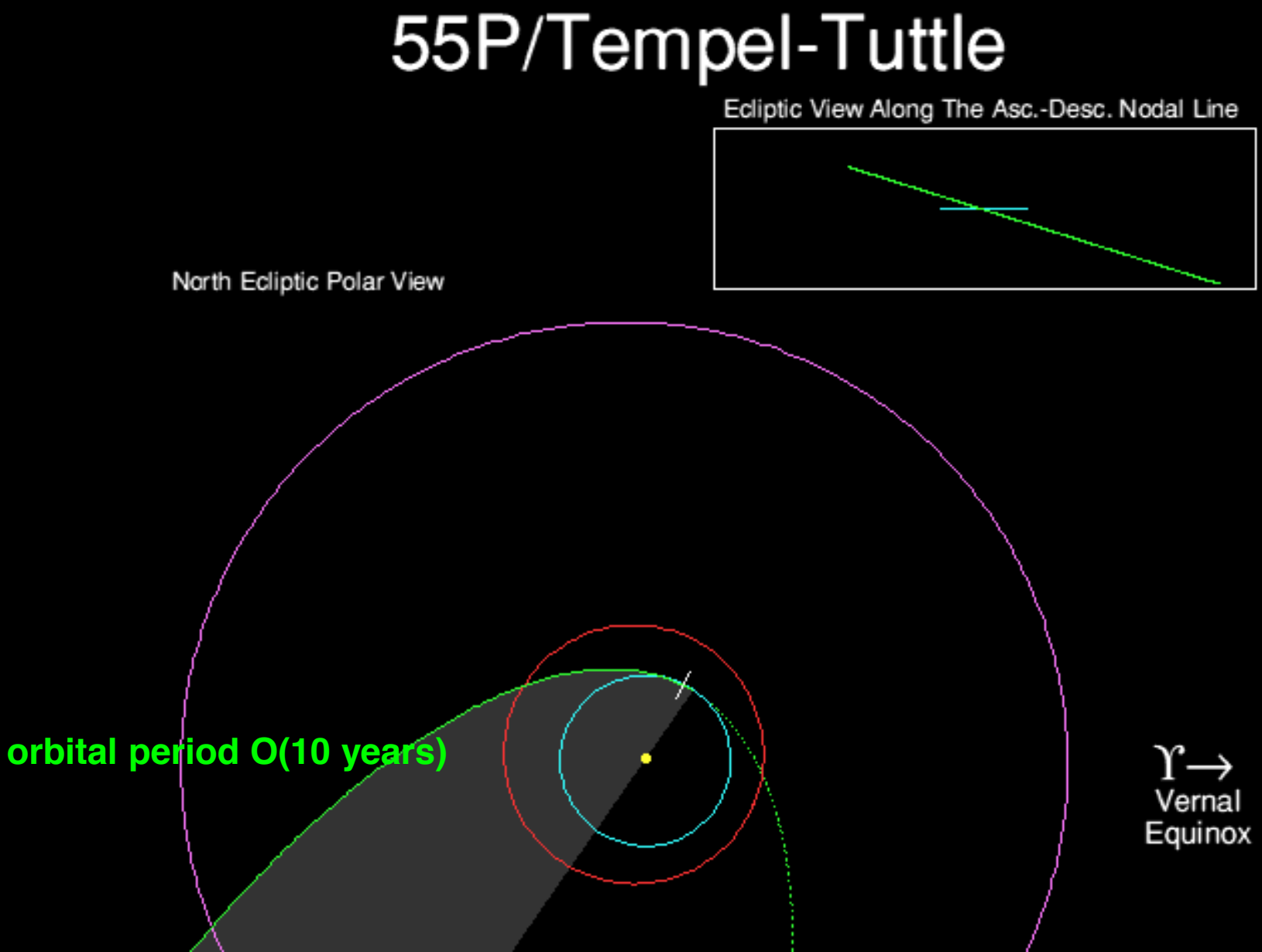
Only a small percentage of comets and asteroids are NEOs

LONG PERIOD COMETS

C/1996 B2 (Hyakutake)

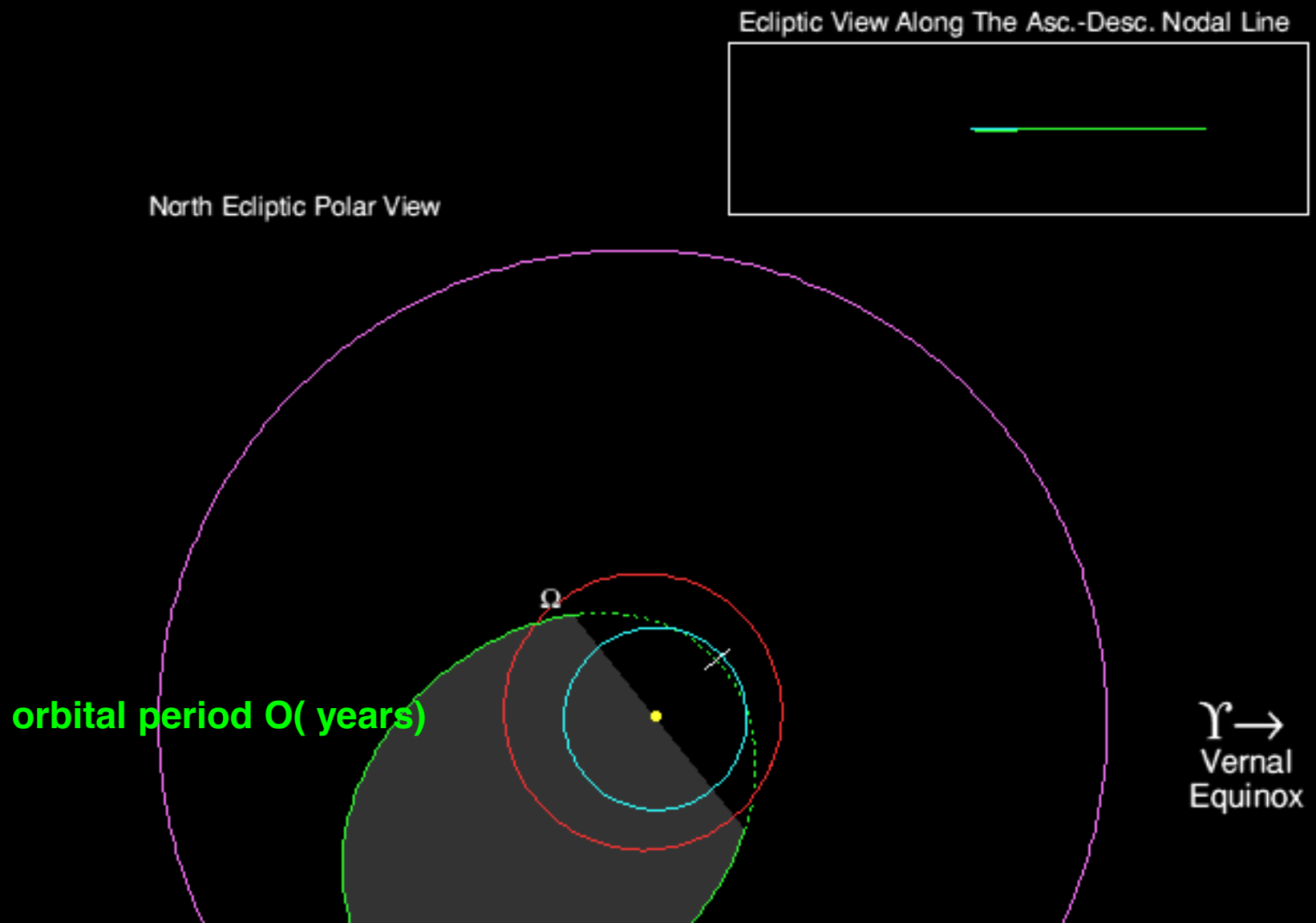


SHORT PERIOD COMETS



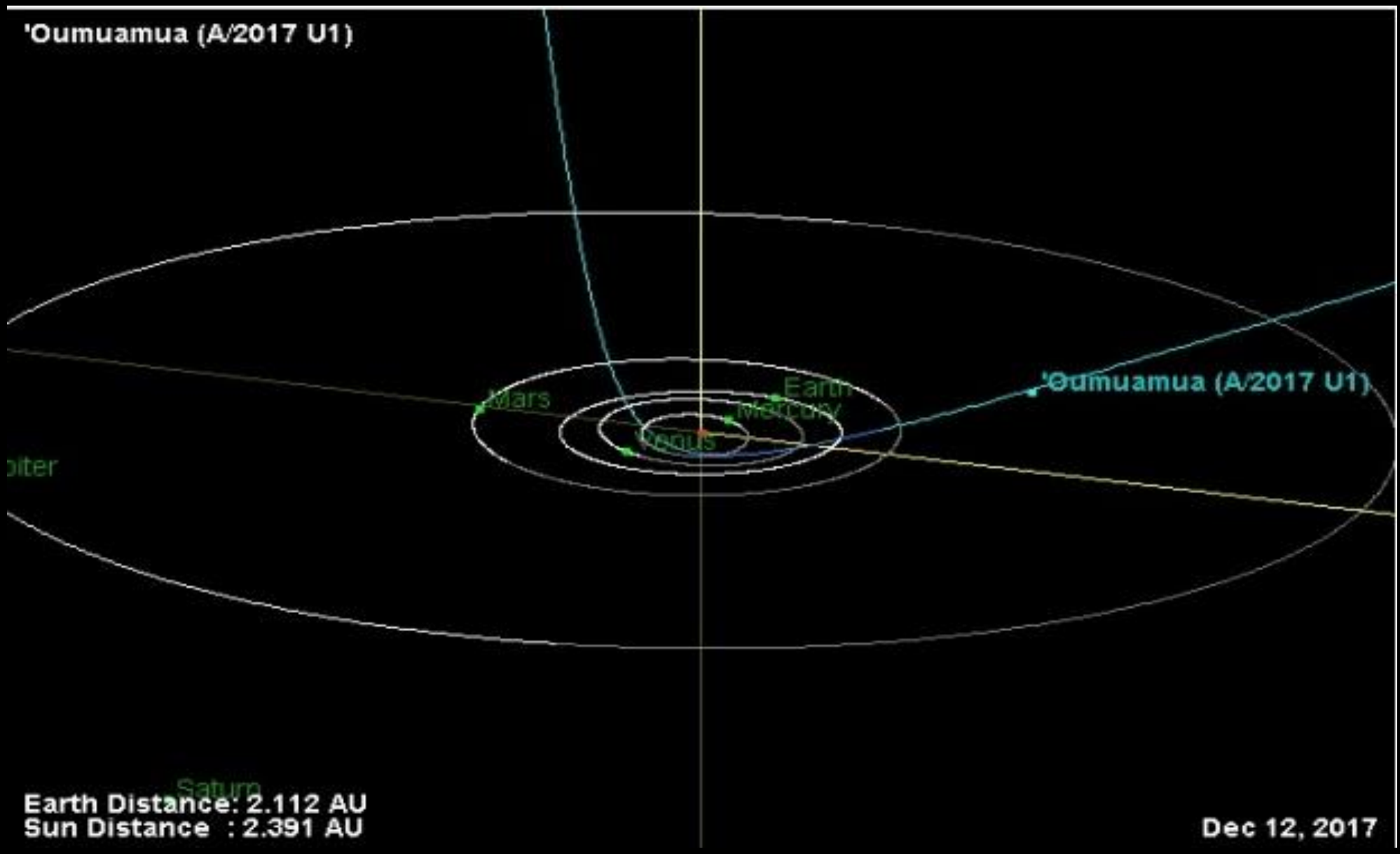
NEAR EARTH ASTEROIDS

(4179) Toutatis



Interstellar UFOs





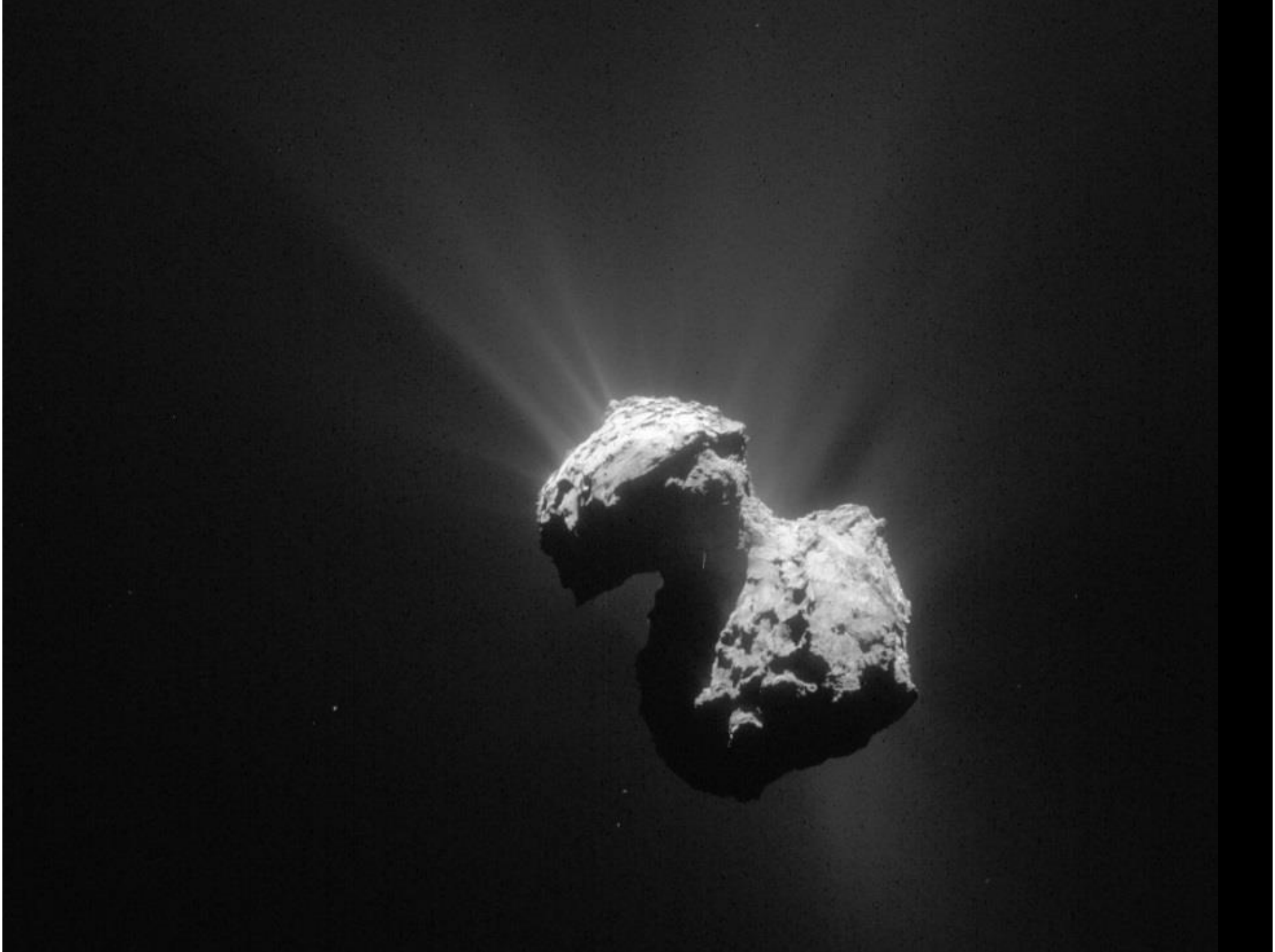


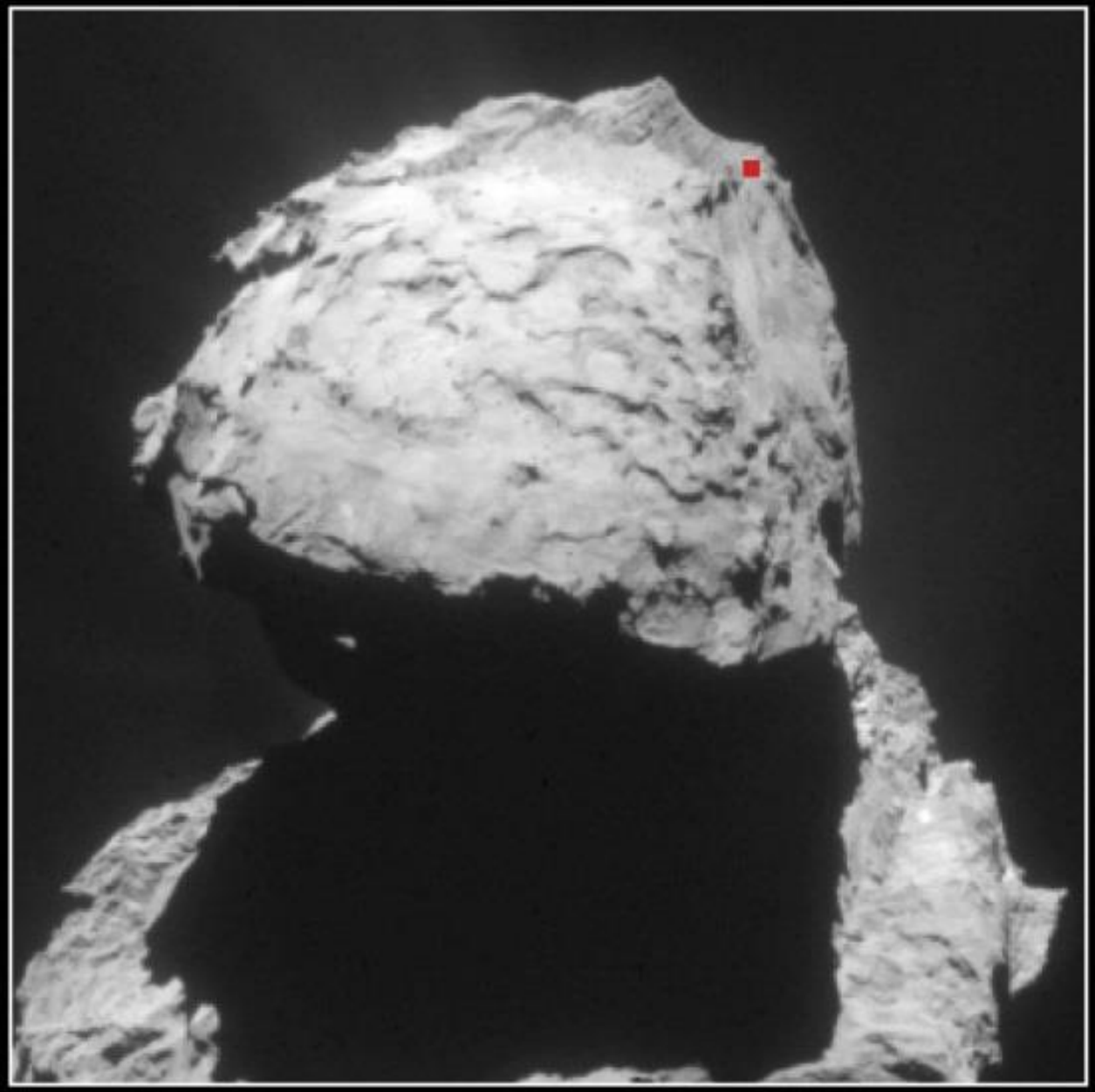


Comets and asteroids are detected from ground by their motion relative to star background




NEAR – 433 Eros
March 07 2000 03:10:00 –27 –158



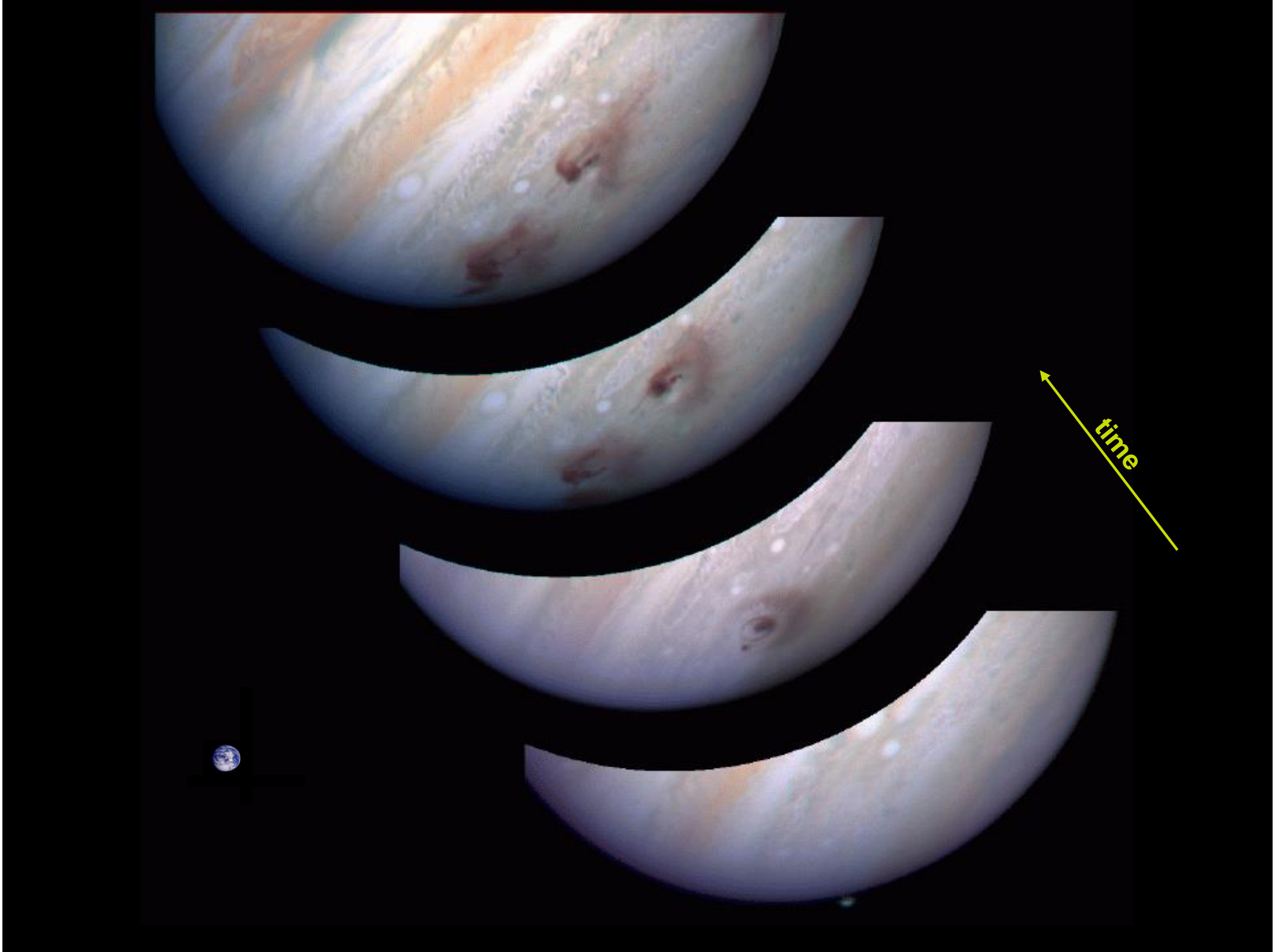


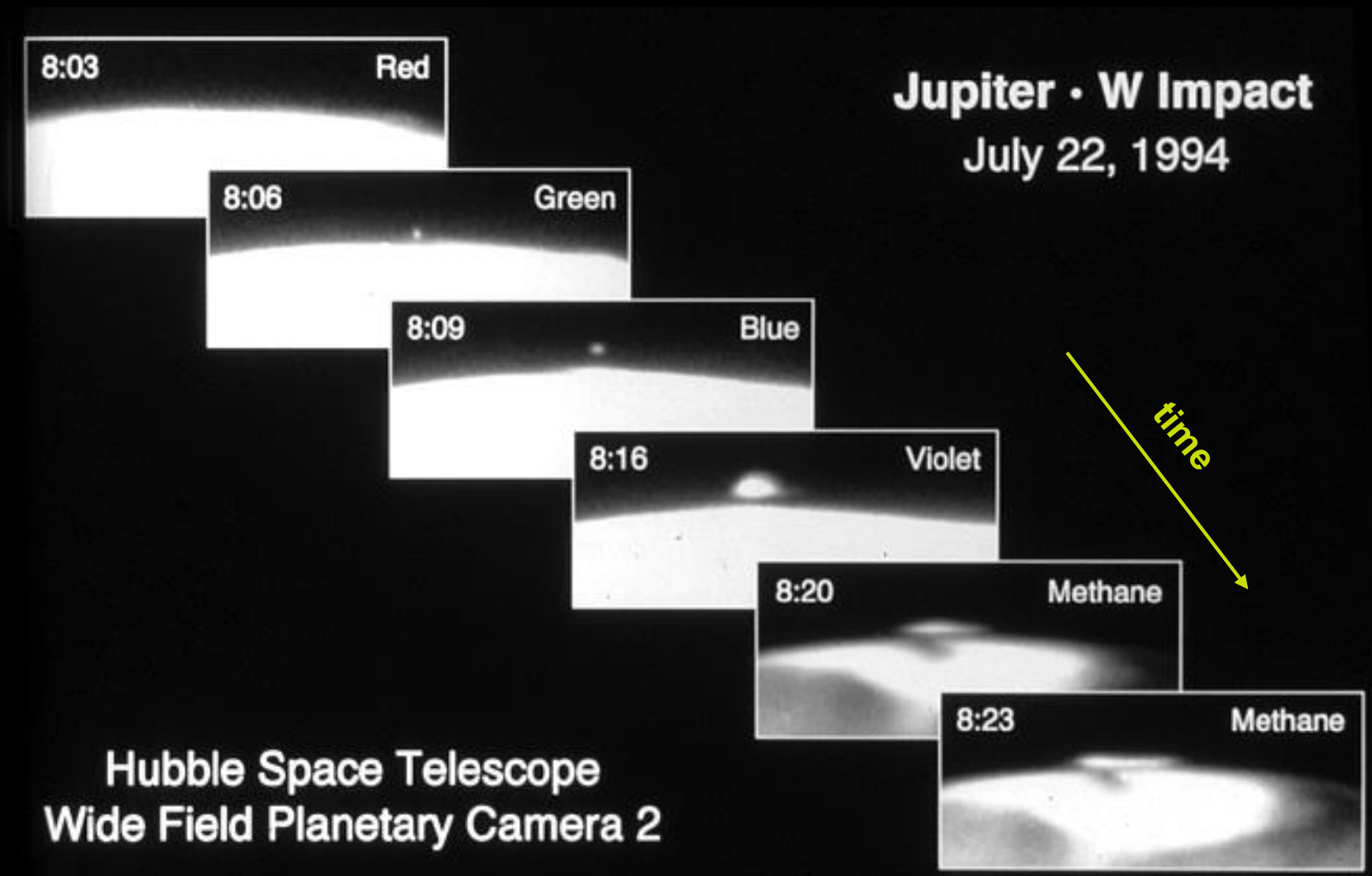


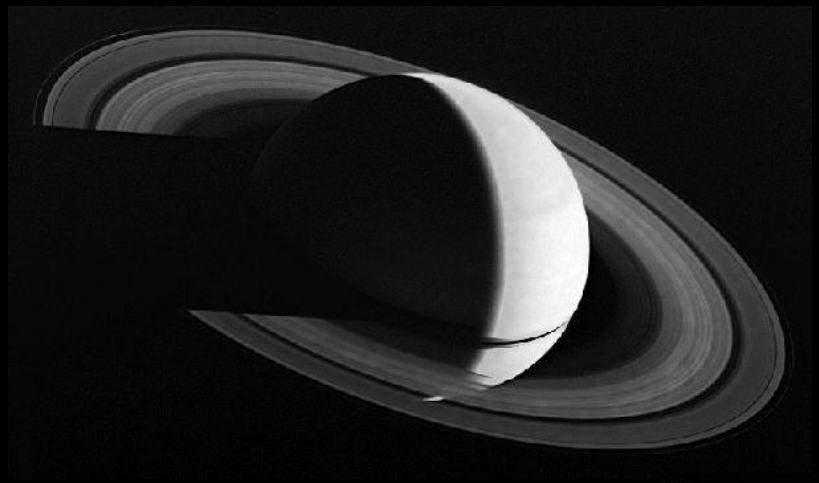


What are the hazards?

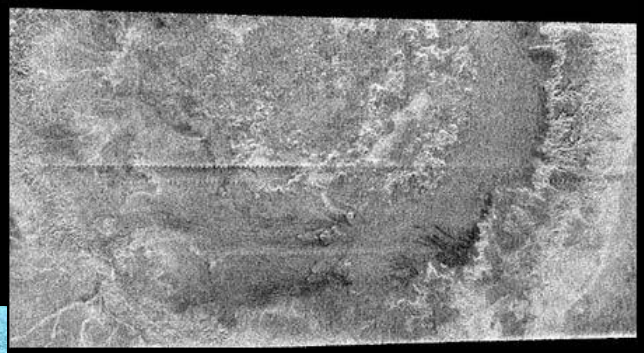


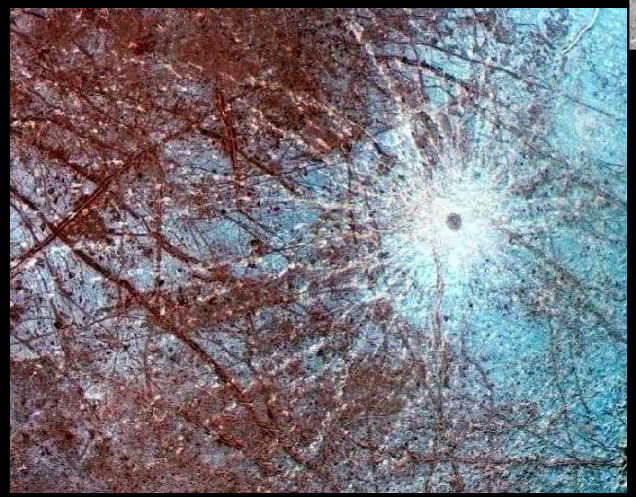

time

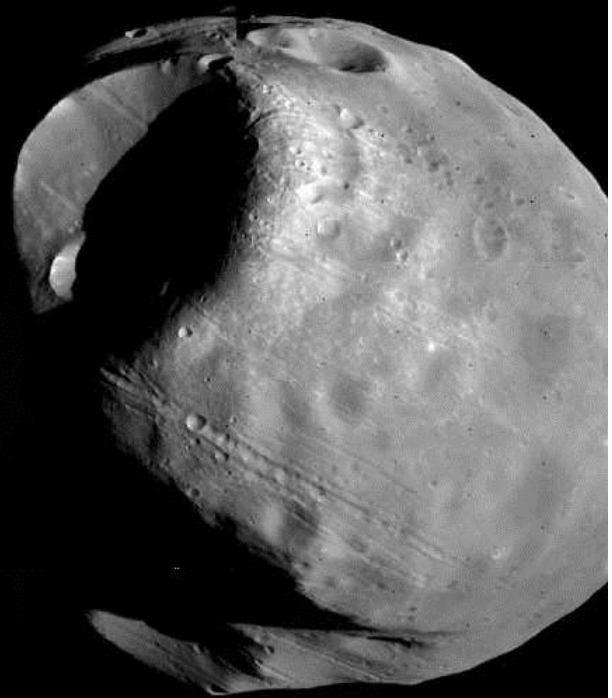

8:03
Red
8:06
Green
8:09
Blue
8:16
Violet
8:20
Methane
8:23
Methane
time
Jupiter · W Impact
July 22, 1994
Hubble Space Telescope
Wide Field Planetary Camera 2









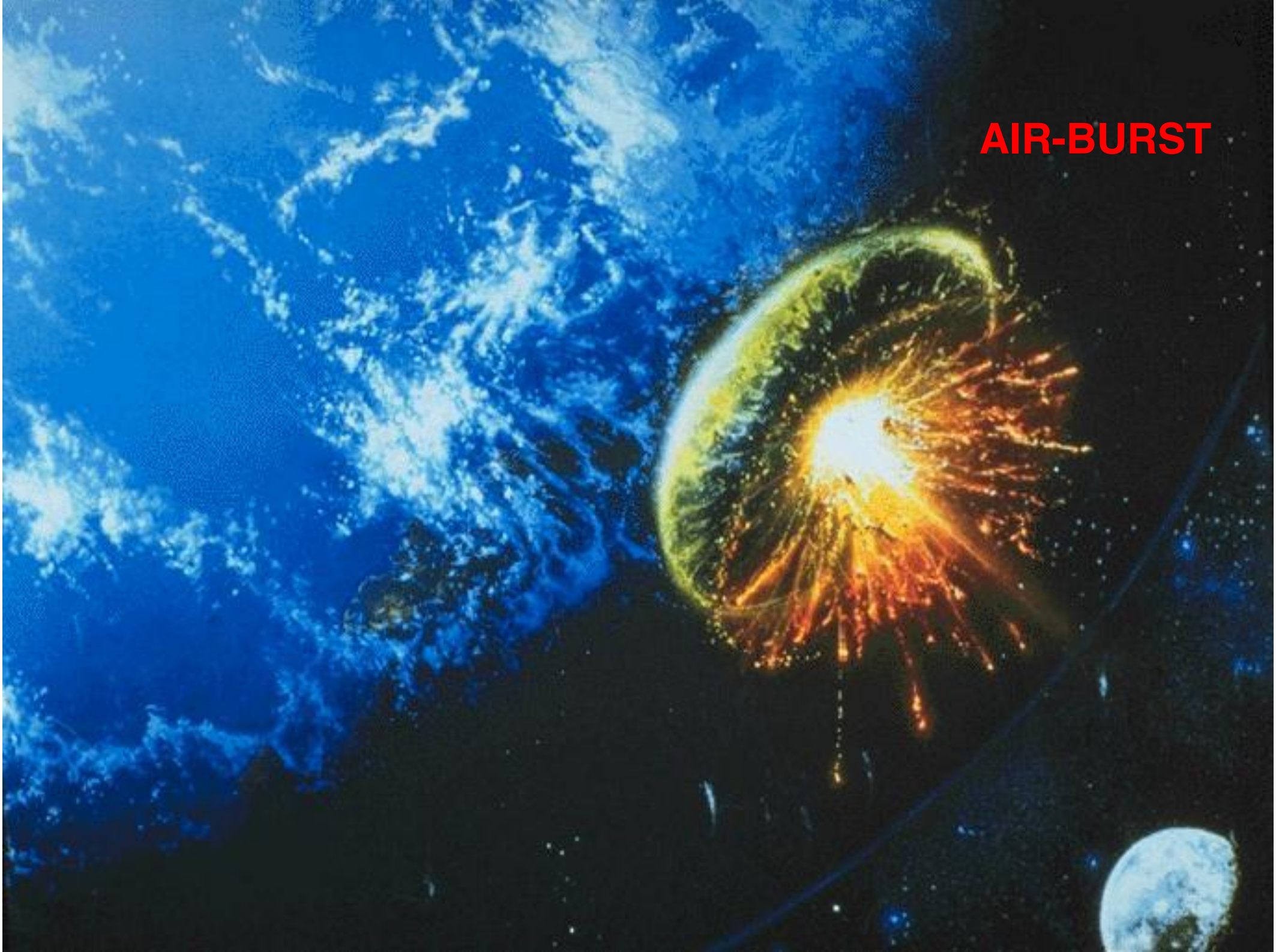


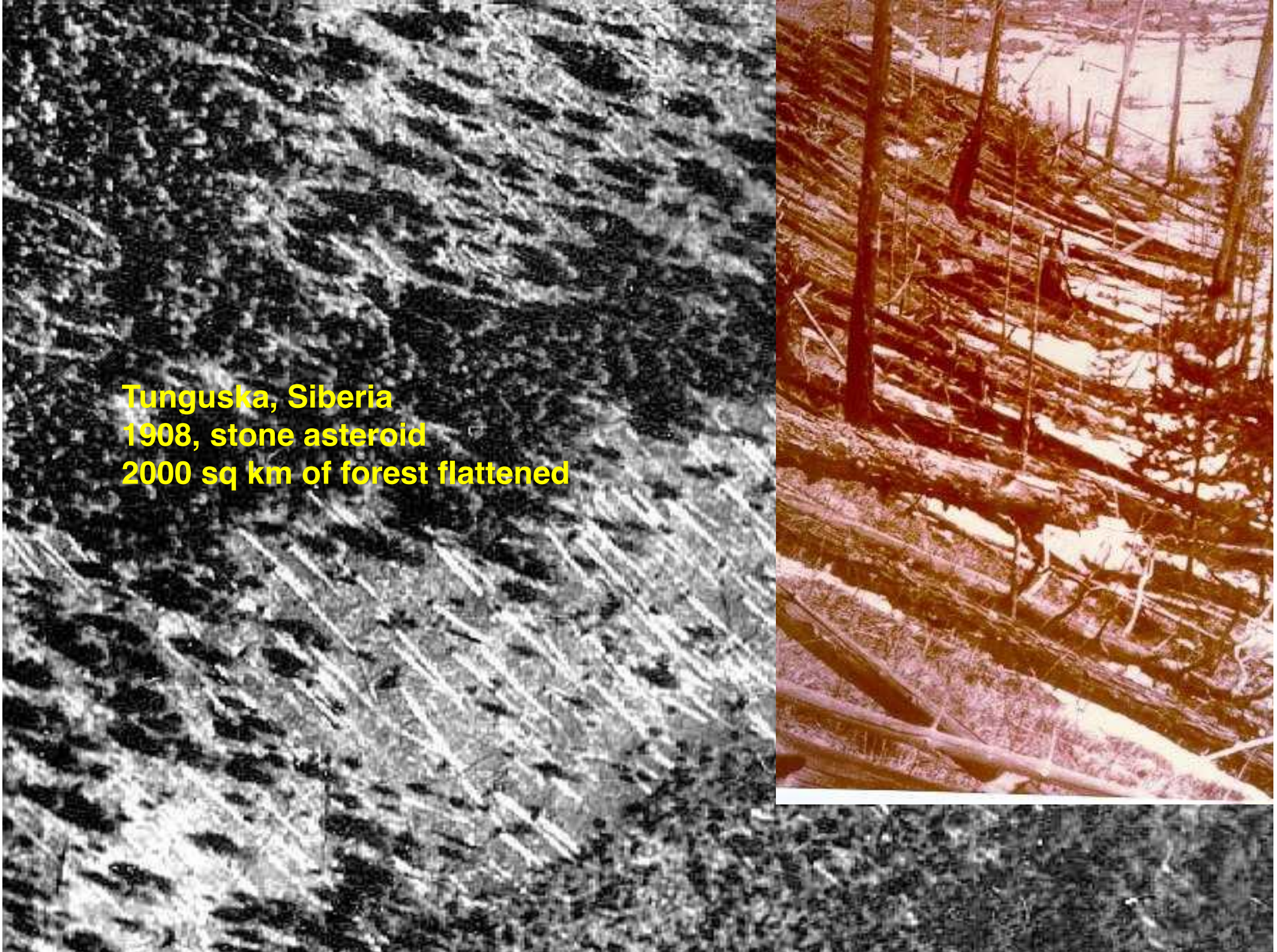
Is the Earth at risk from NEOs?








AIR-BURST

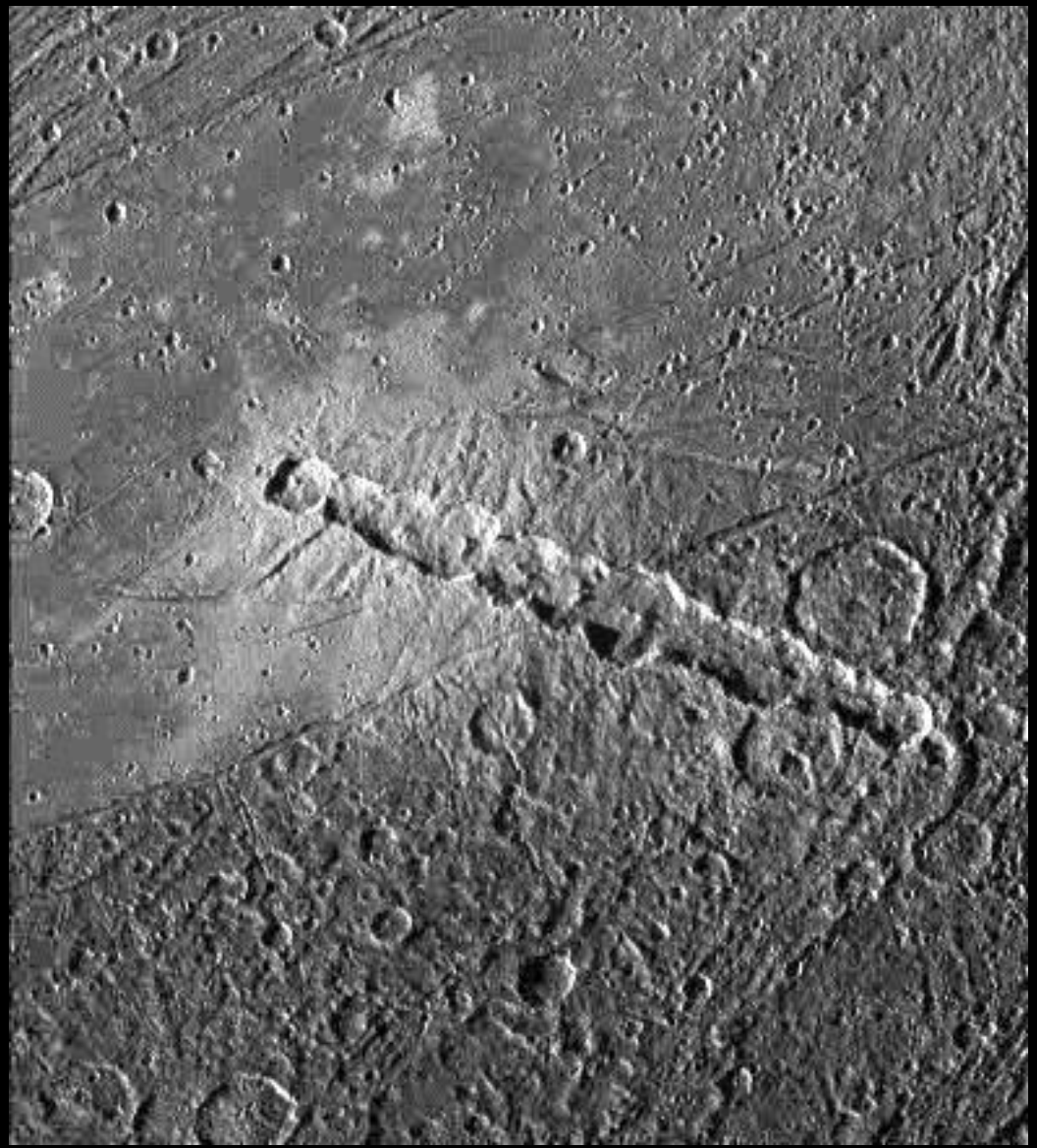

Tunguska, Siberia
1908, stone asteroid
2000 sq km of forest flattened


Barringer Crater, Arizona
49,000 years ago
nickel-iron asteroid



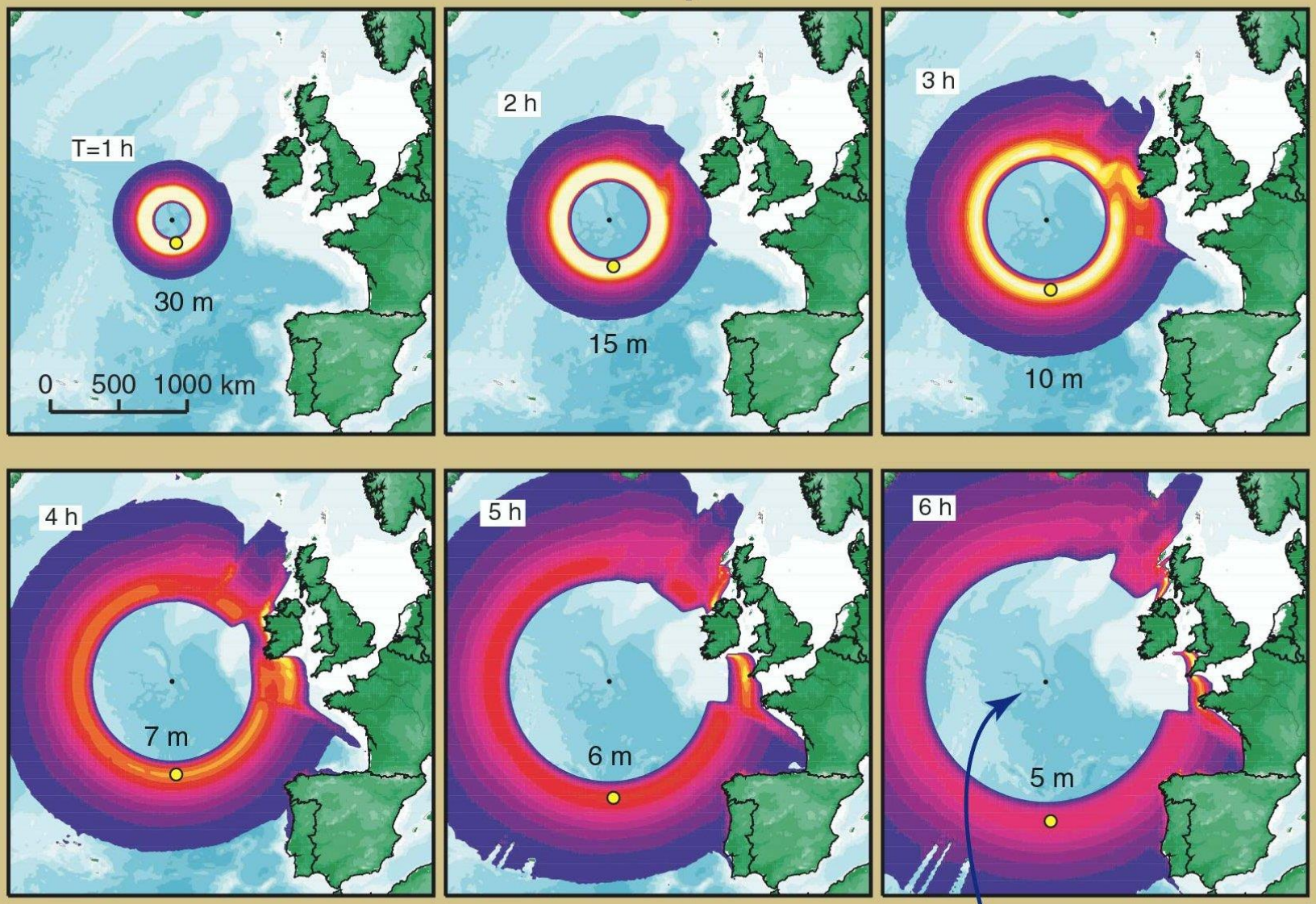


Multiple impacts are common (e.g. Moon and Gannymede)





North Atlantic Impact Tsunami

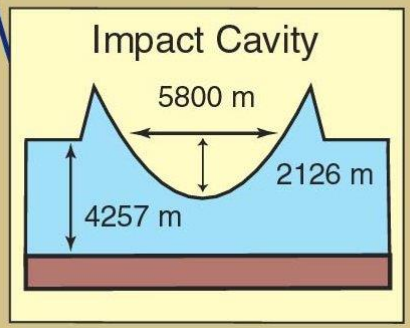


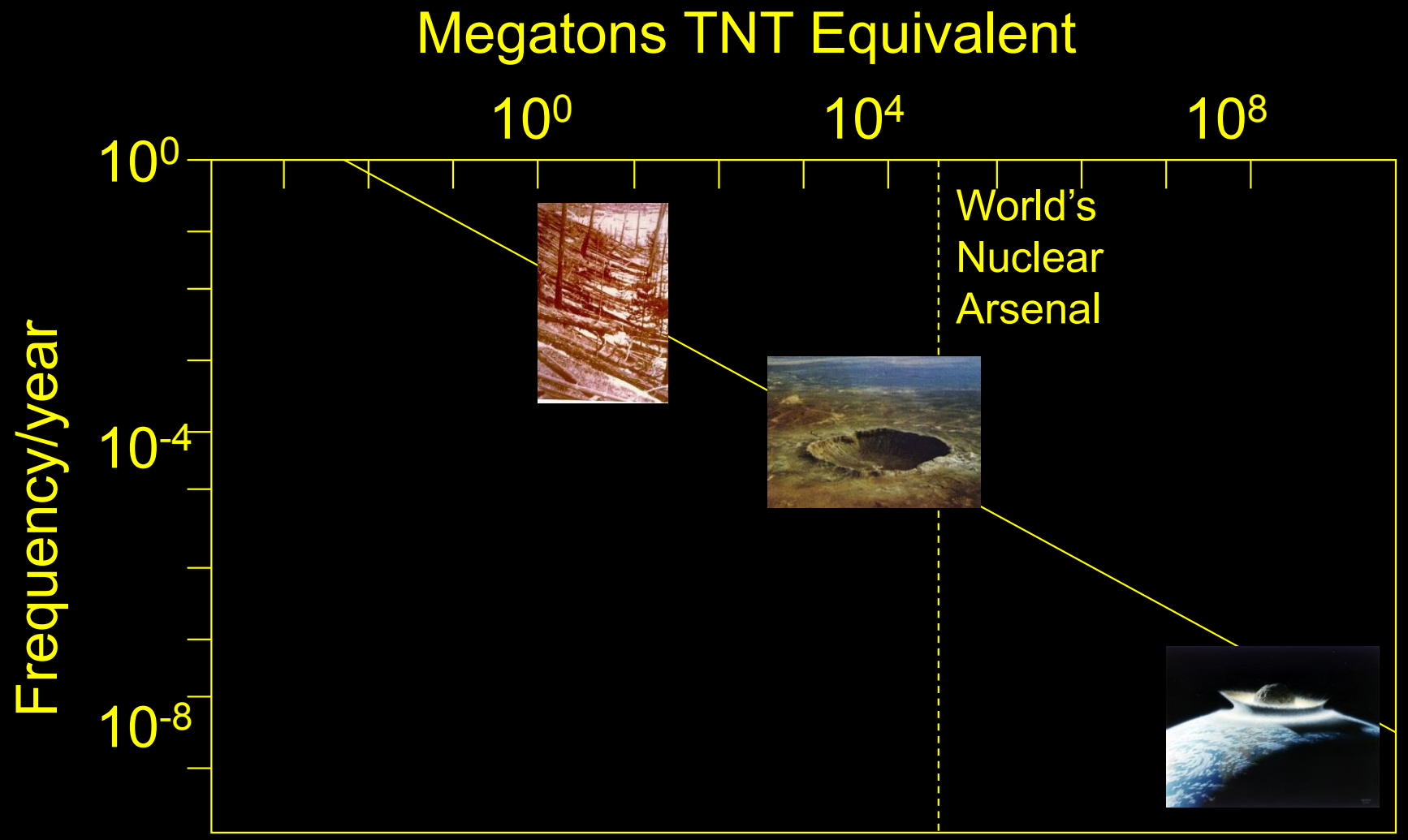
Impactor Diameter= 200 m Water Depth at Impact= 4257 m

Tsunami Energy= 2.8×10^{17} J T_{min}= 50s

1.5 3.0 4.5 6.0 7.5 9.0 >9.75

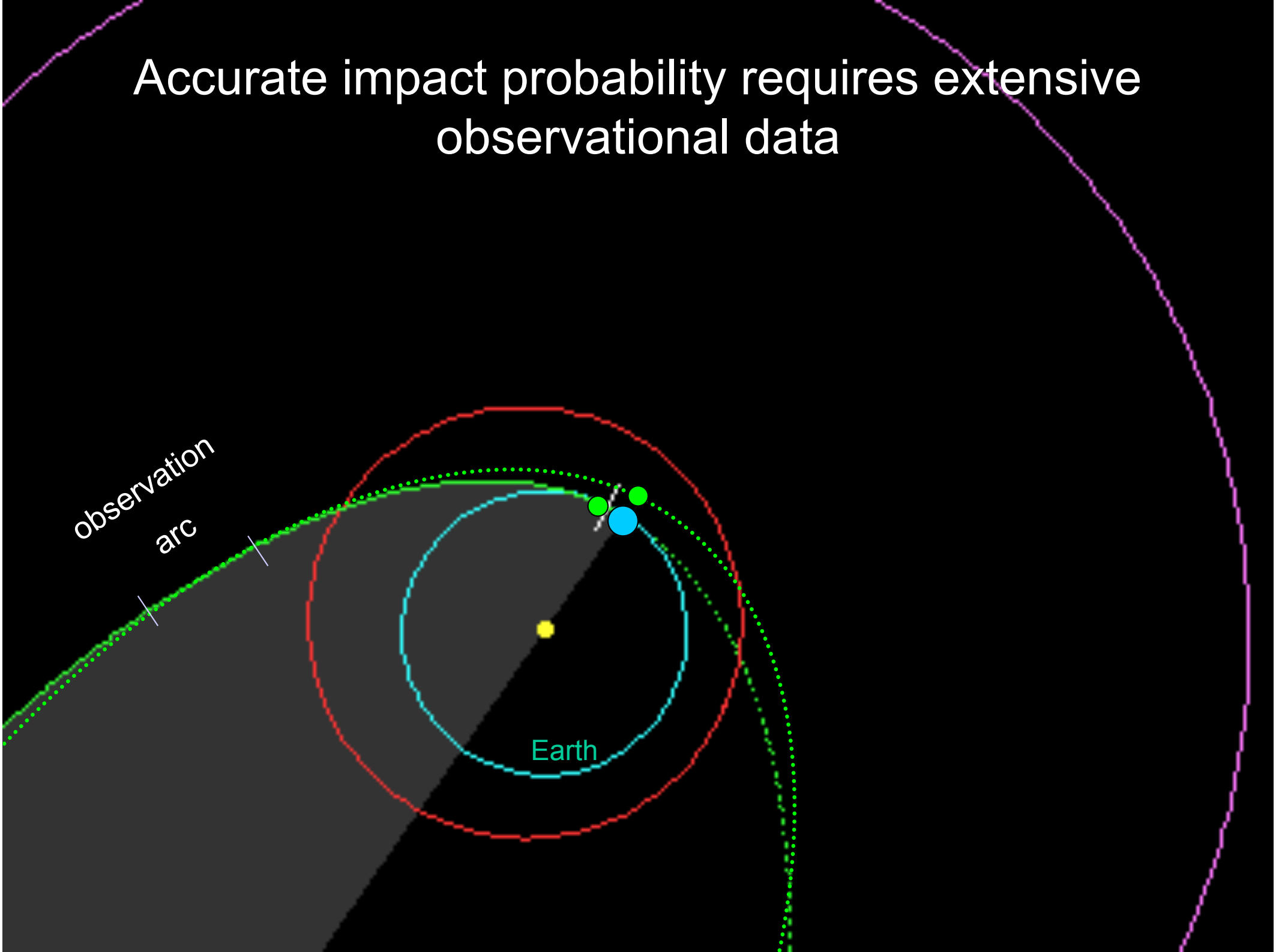
Tsunami Envelope Height in Meters



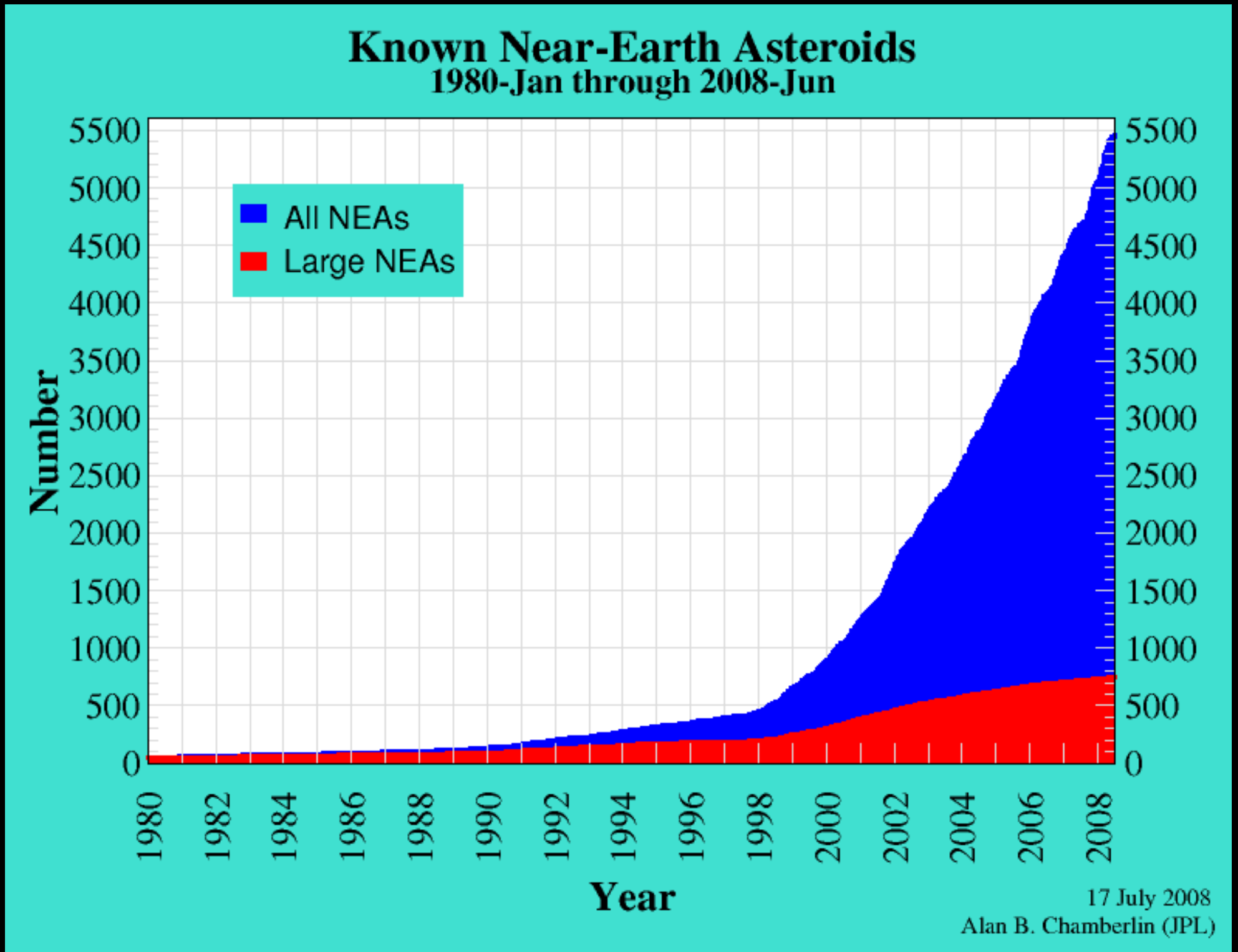


Major uncertainty is frequency, small data set

Accurate impact probability requires extensive observational data




Carancas, Peru
2007 impact crater


Known Near-Earth Asteroids
1980-Jan through 2008-Jun
All NEAs
Large NEAs
Number
5500
5000
4500
4000
3500
3000
2500
2000
1500
1000
500
0
1980
1982
1984
1986
1988
1990
1992
1994
1996
1998
2000
2002
2004
2006
2008
Year
17 July 2008
Alan B. Chamberlin (JPL)

NEO risk management issues to be addressed



OBJECT IDENTIFICATION	ORBIT DETERMINATION	CONSEQUENCE DETERMINATION	IN-SITU CHARACTERISATION	MITIGATION	EVALUATION
POLICY					
CATEGORISING, CATALOGUING, NOTIFICATION	SENSOR TASKING	NOTIFICATION, RISK CRITERIA	ROLES, RESPONSIBILITIES	TREATY COMPLIANCE, AUTHORITY, RESPONSIBILITY	OUTCOME NOTIFICATION
INFRASTRUCTURE					
SENSOR AVAILABILITY, LOCATION	SENSOR COORDINATION, PRIORITISATION	TOOLS, VULNERABILITY	PAYLOAD DELIVERY, DATA COMMUNICATION	PAYLOAD DELIVERY, COMMAND & CONTROL	SENSOR TASKING
SCIENCE					
SENSOR LOCATION, DETECTION THRESHOLD	FOLLOW -UP STRATEGY	MULTI-DISCIPLINARY STUDIES	SENSOR DEVELOPMENT, DATA EXPLOITATION	NEGATION PHYSICS	SHORT TERM TRAJECTORY PREDICTION


VISITORS LOUNGE



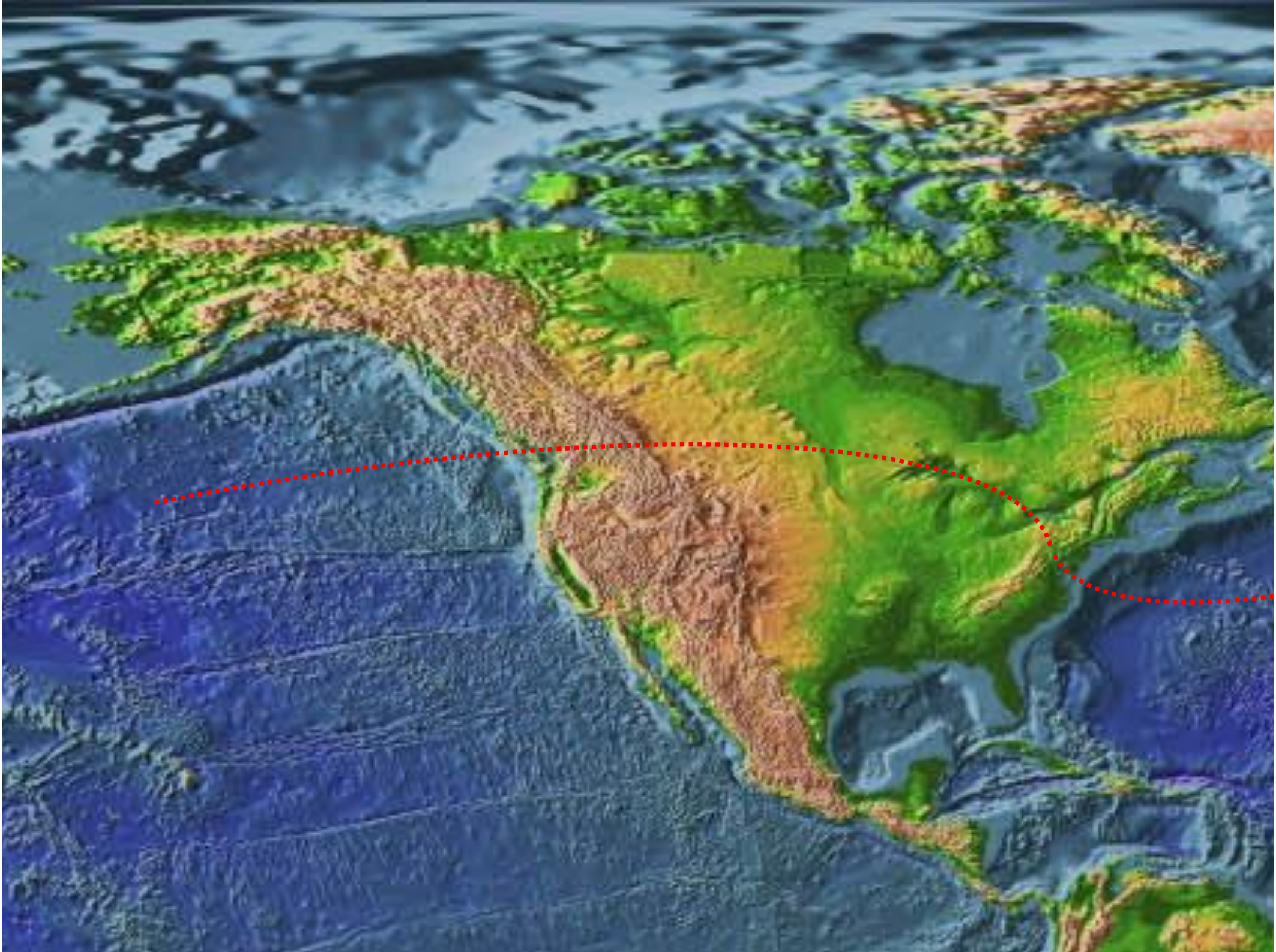



THAILAND
UKRAINE


UNITED STATES







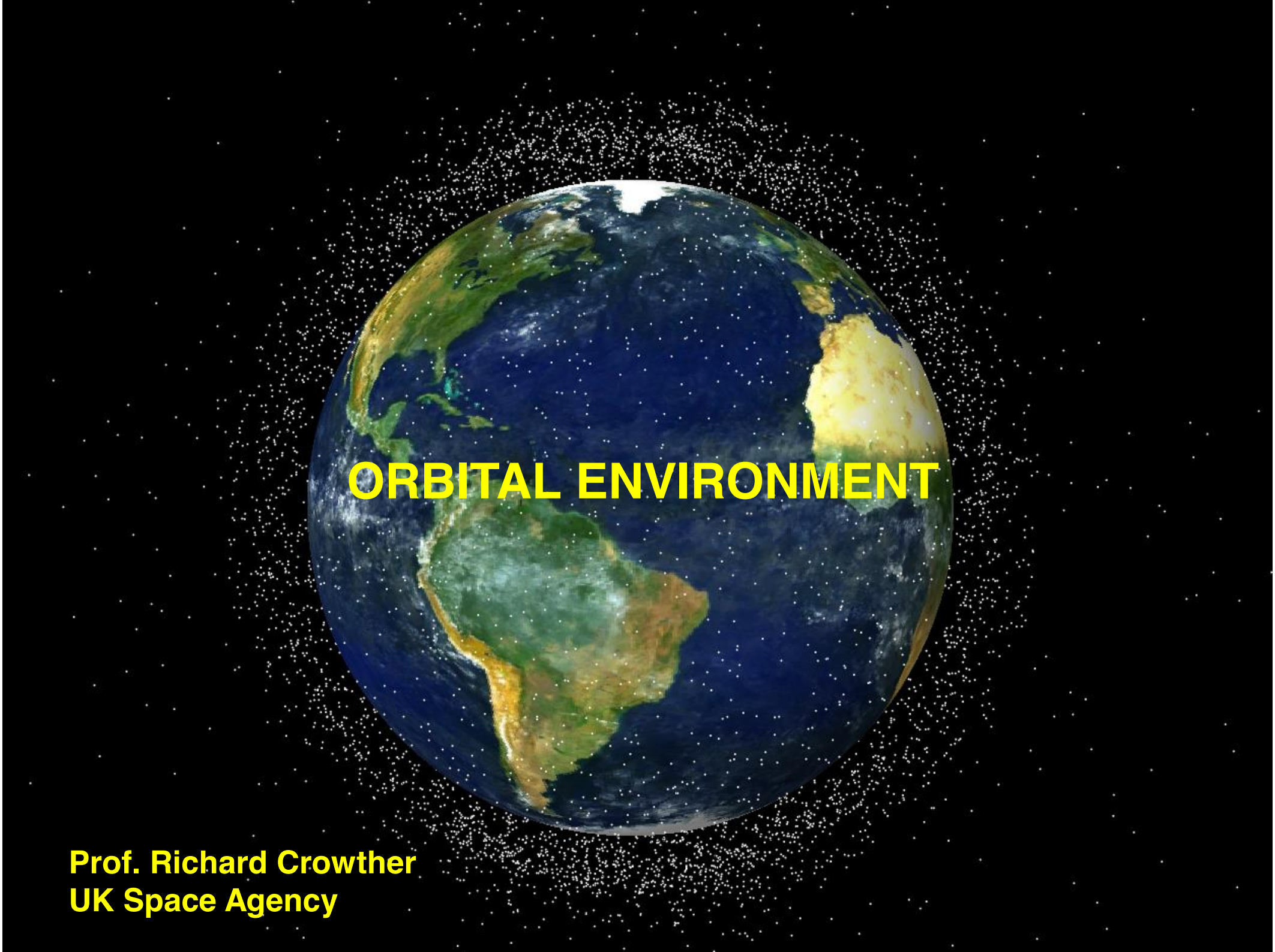
International Asteroid Warning Network




International Asteroid Warning Network
Space Mission Planning Advisory Group

Asteroid Day





Prof. Richard Crowther
UK Space Agency

Outline

- What is in orbit around the Earth?
- How much space debris is there?
- What is the future for the space environment?
- What is the solution to space debris?

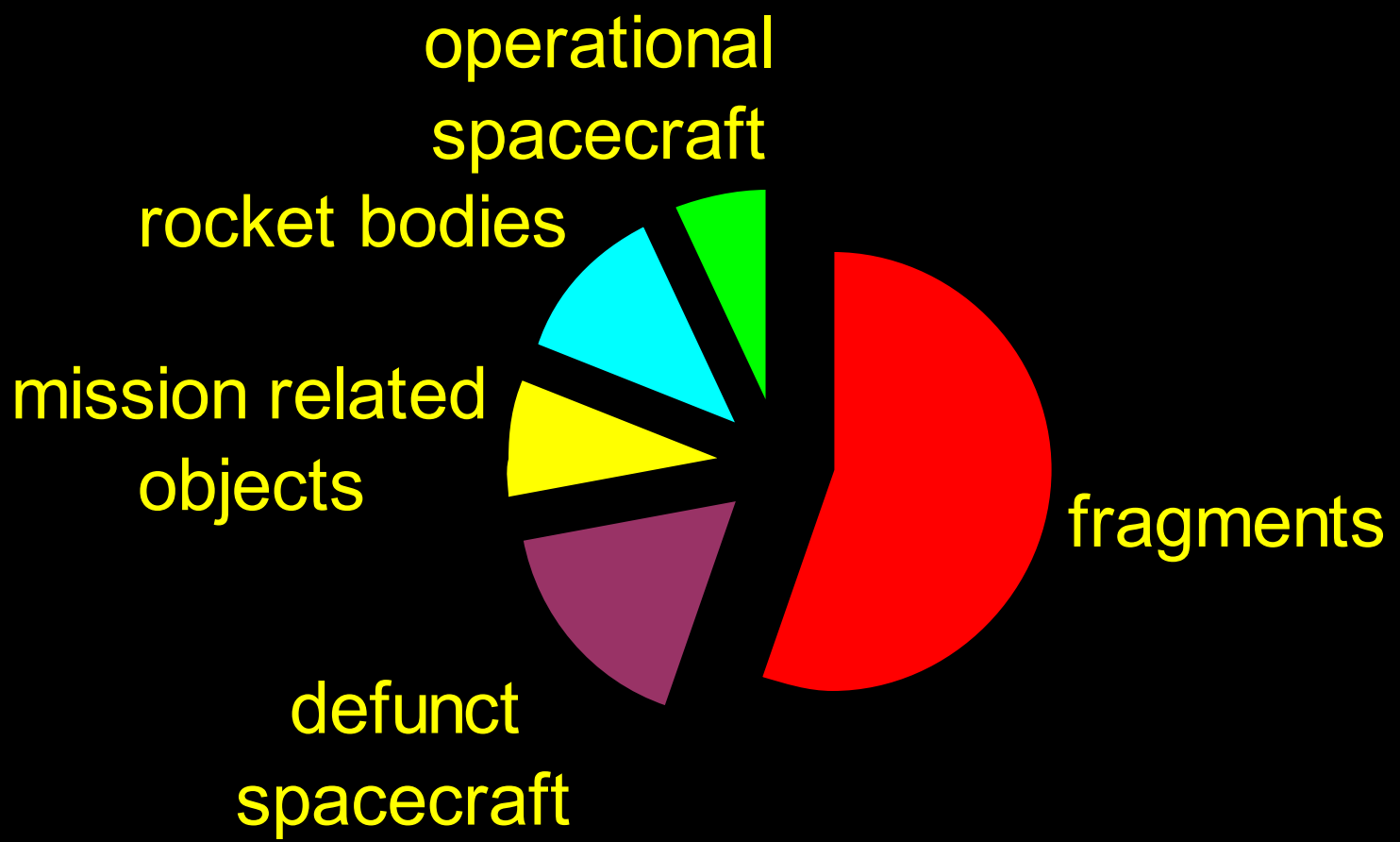
WHAT IS IN ORBIT AROUND THE EARTH?



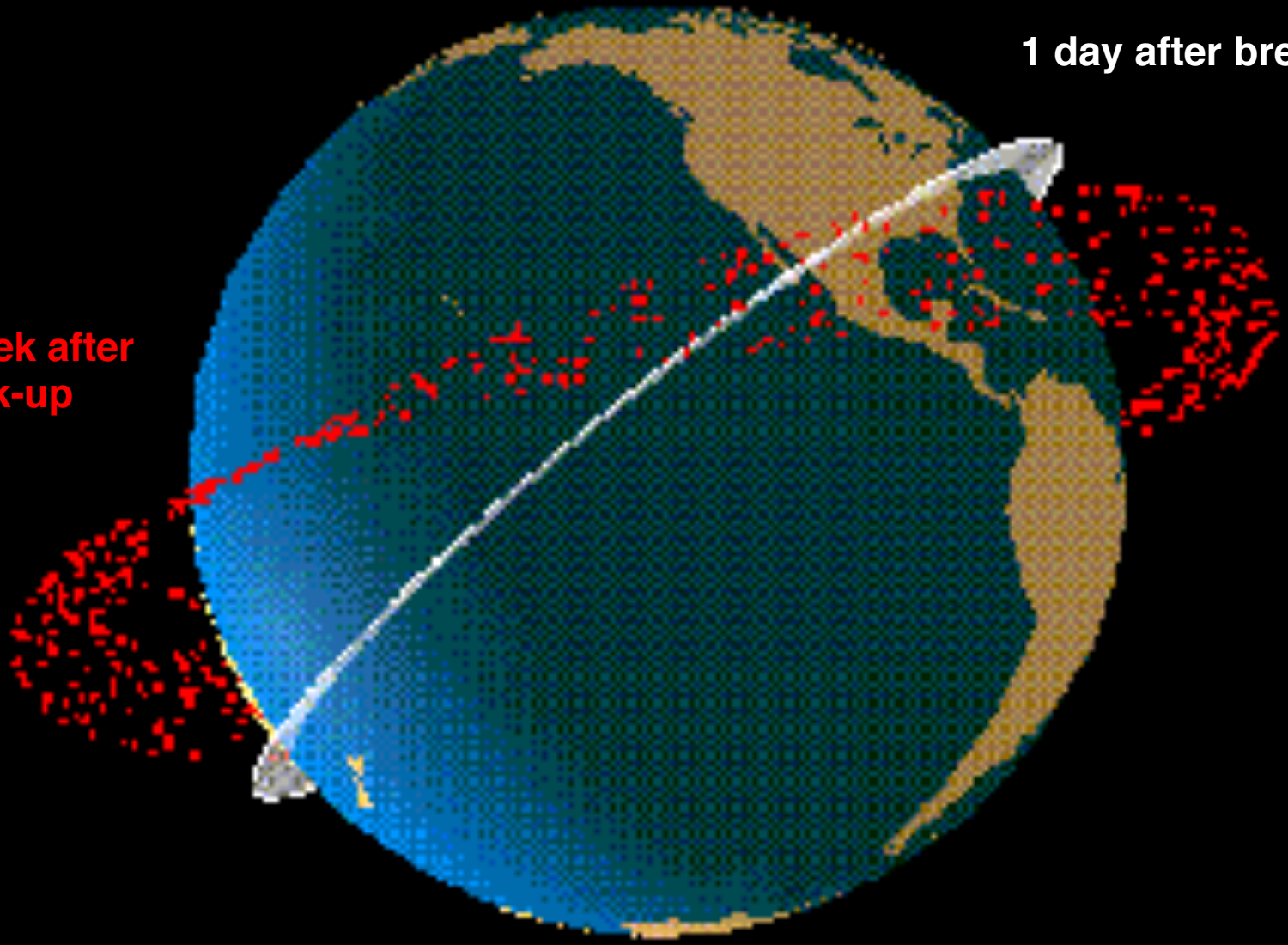
near-Earth satellite population reflects use of space

>16000 catalogued objects concentrated in distinct orbits with unique characteristics

CATEGORIES OF CATALOGUED OBJECTS



FRAGMENTS FROM BREAK-UP QUICKLY DISPERSE



HOW MUCH SPACE DEBRIS IS THERE?

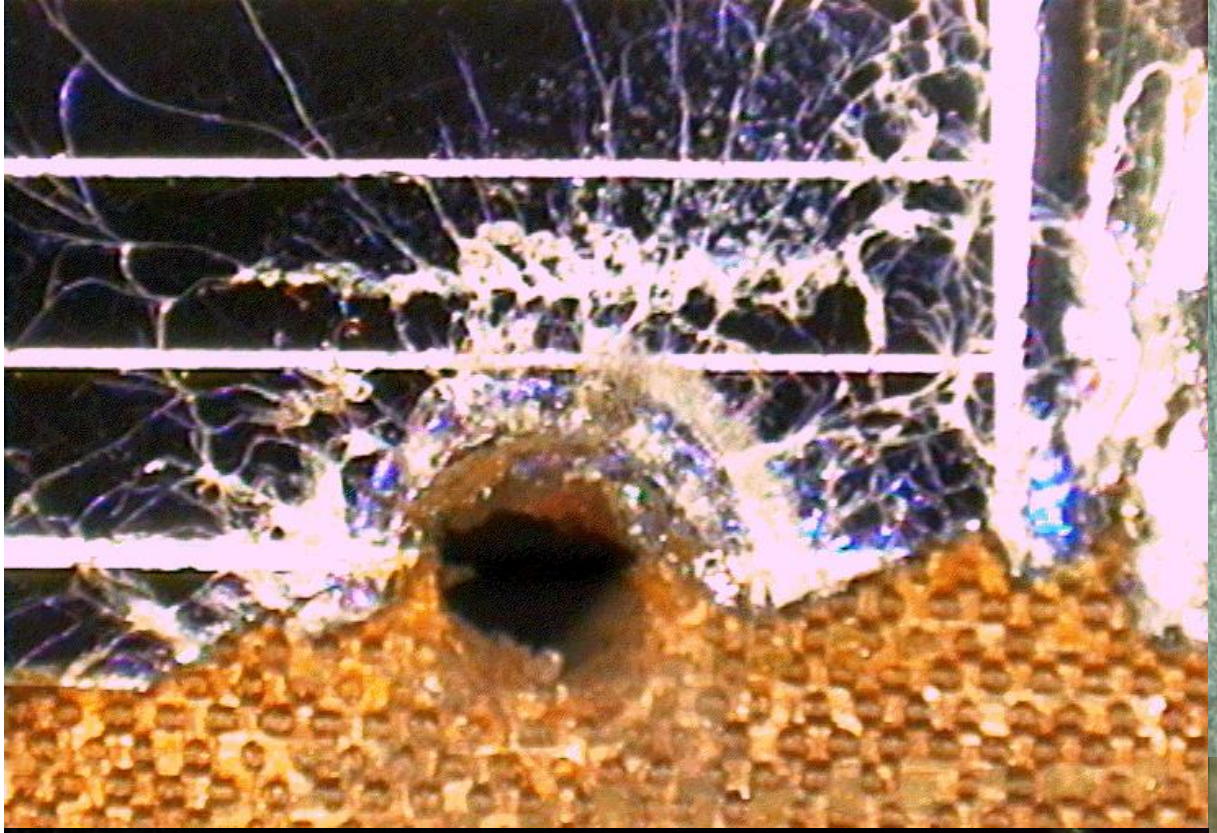
Estimating the debris population



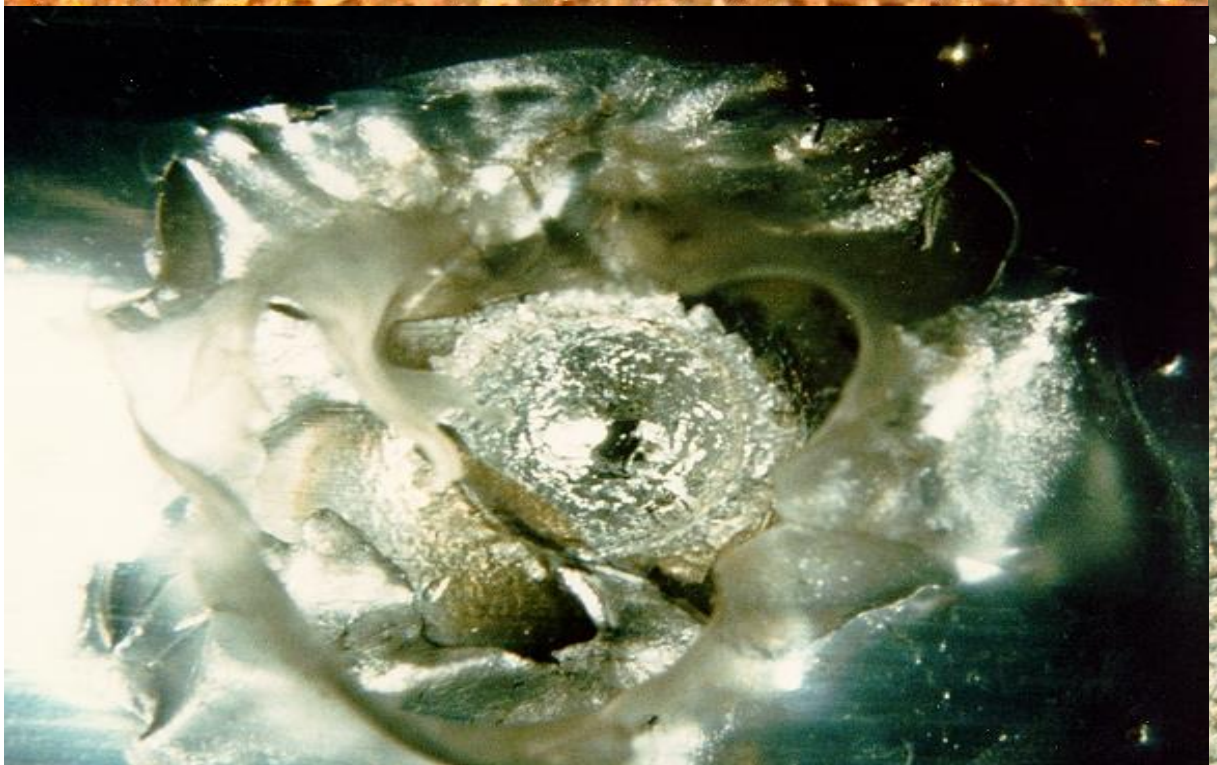


BAY 5


T180-1
T180-2
T180-3
T180-4
T180-5
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T180-9
T180-10
T180-11
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T180-63
T180-64
T180-65
T180-67
T180-68
T180-69
-V3
150
180









Estimated Debris Population

Size	Number	% Mass
>10 cm	>20,000	99.93
1-10 cm	>500,000	0.035
<1 cm	>50,000,000	0.035
Total	>50,000,000	>6,000 tonnes

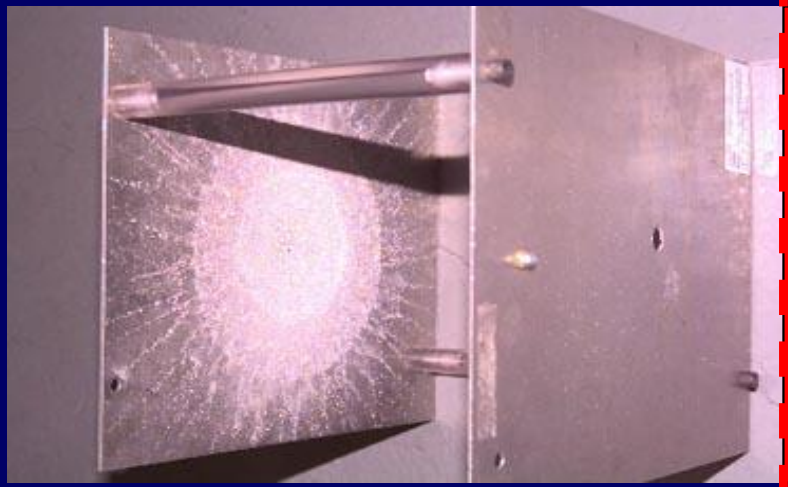
Estimated Debris Population

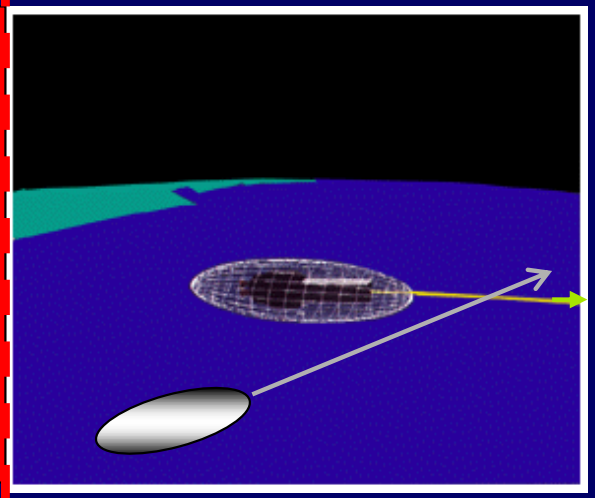
Size	Number	% Mass
>10 cm	>20,000	99.83
1-10 cm	>2,000,000	0.135
<1 cm	>50,000,000	0.035
Total	>50,000,000	>6,000 tonnes

Mind the gap!

SHIELDING

COLLISION AVOIDANCE

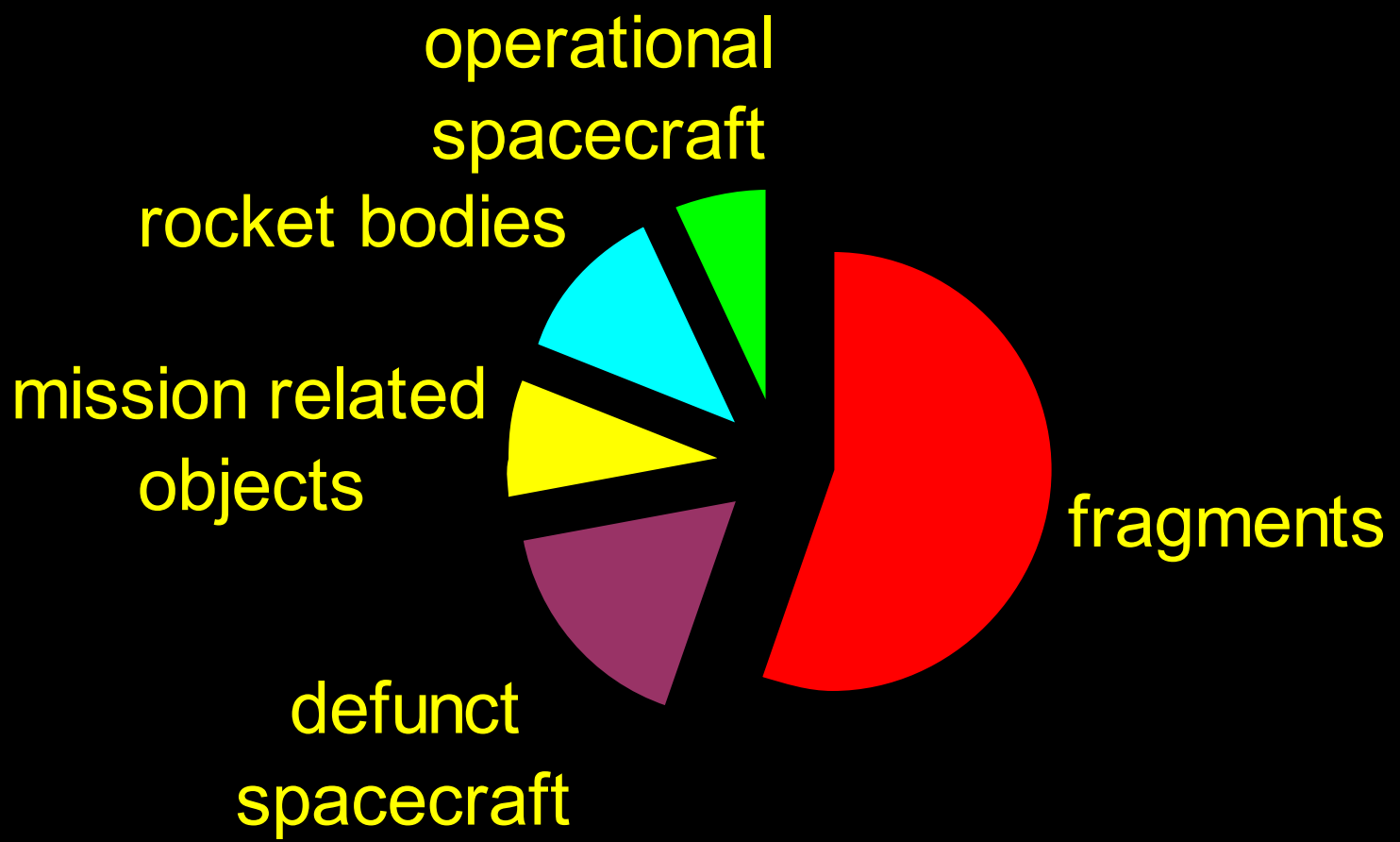




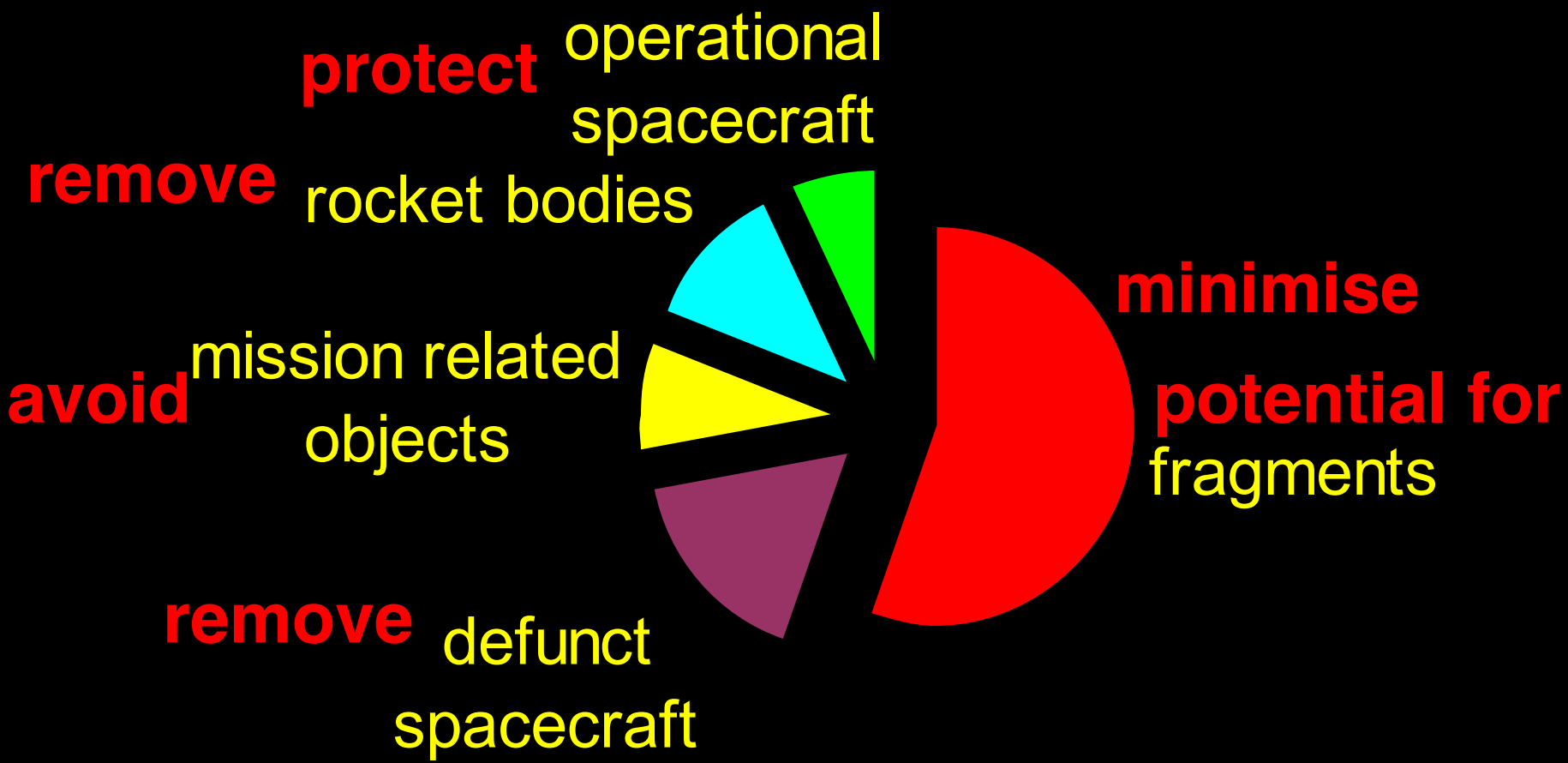
1μm 10μm 100μm 1mm 1cm 10cm 1m 10m

WHAT IS THE SOLUTION TO SPACE DEBRIS?

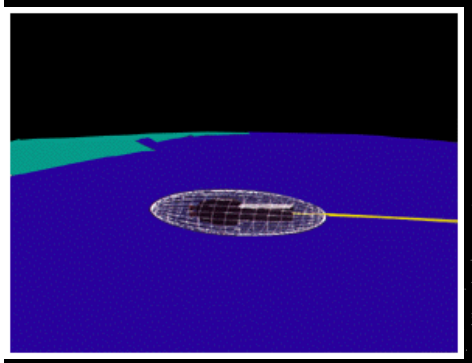
MITIGATION OBJECTIVES

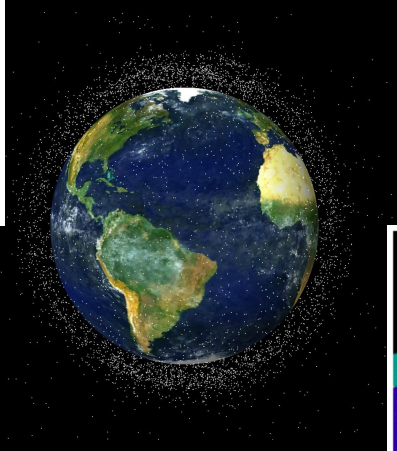


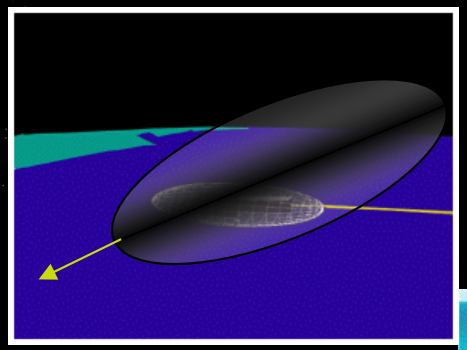
MITIGATION OBJECTIVES



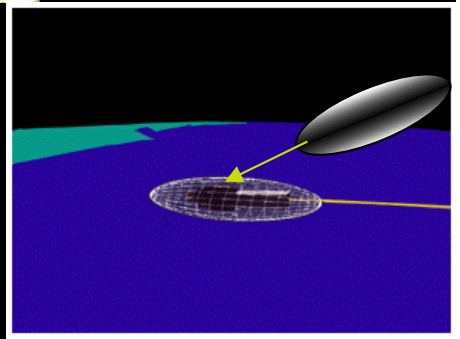
Collision Avoidance

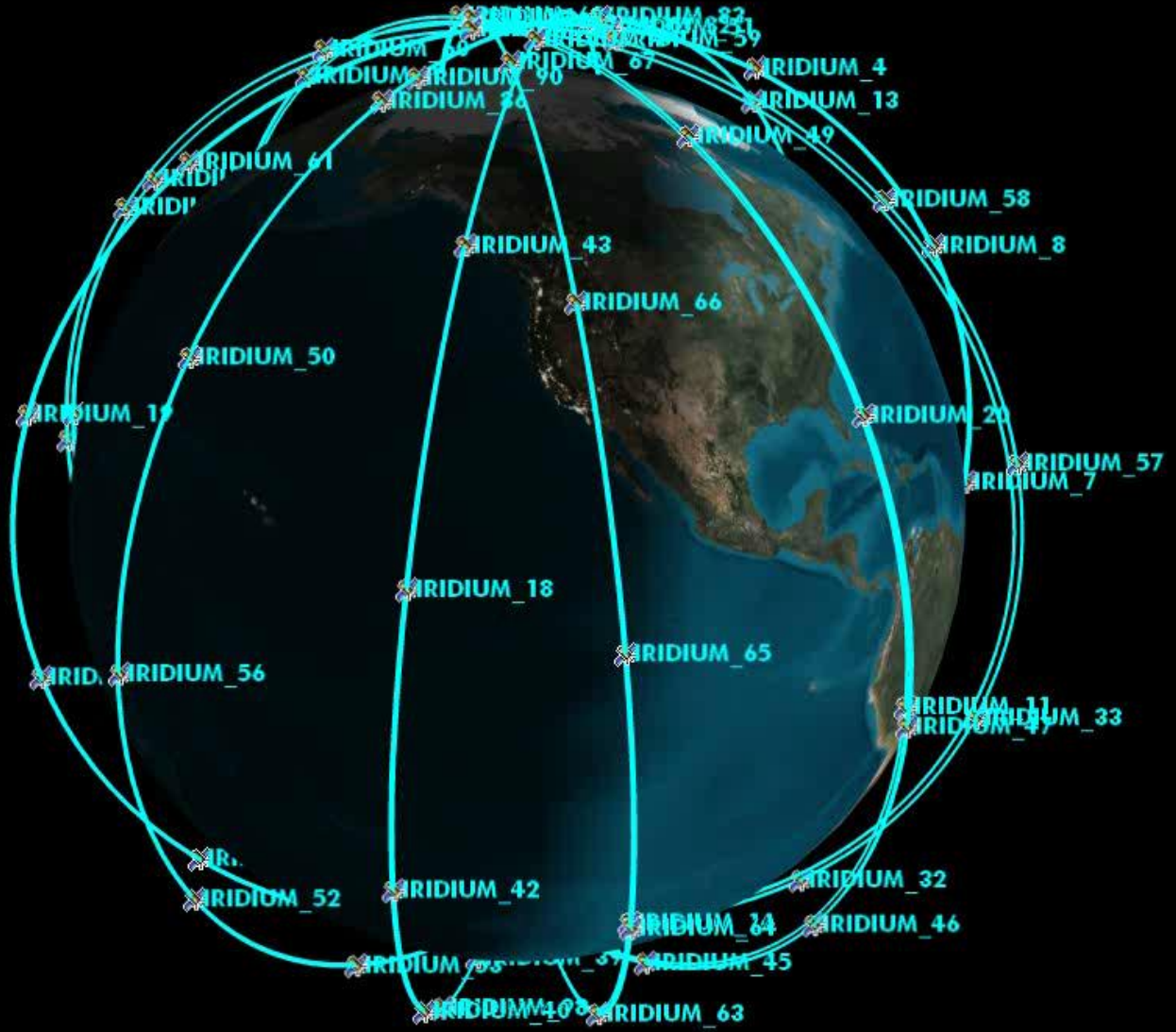


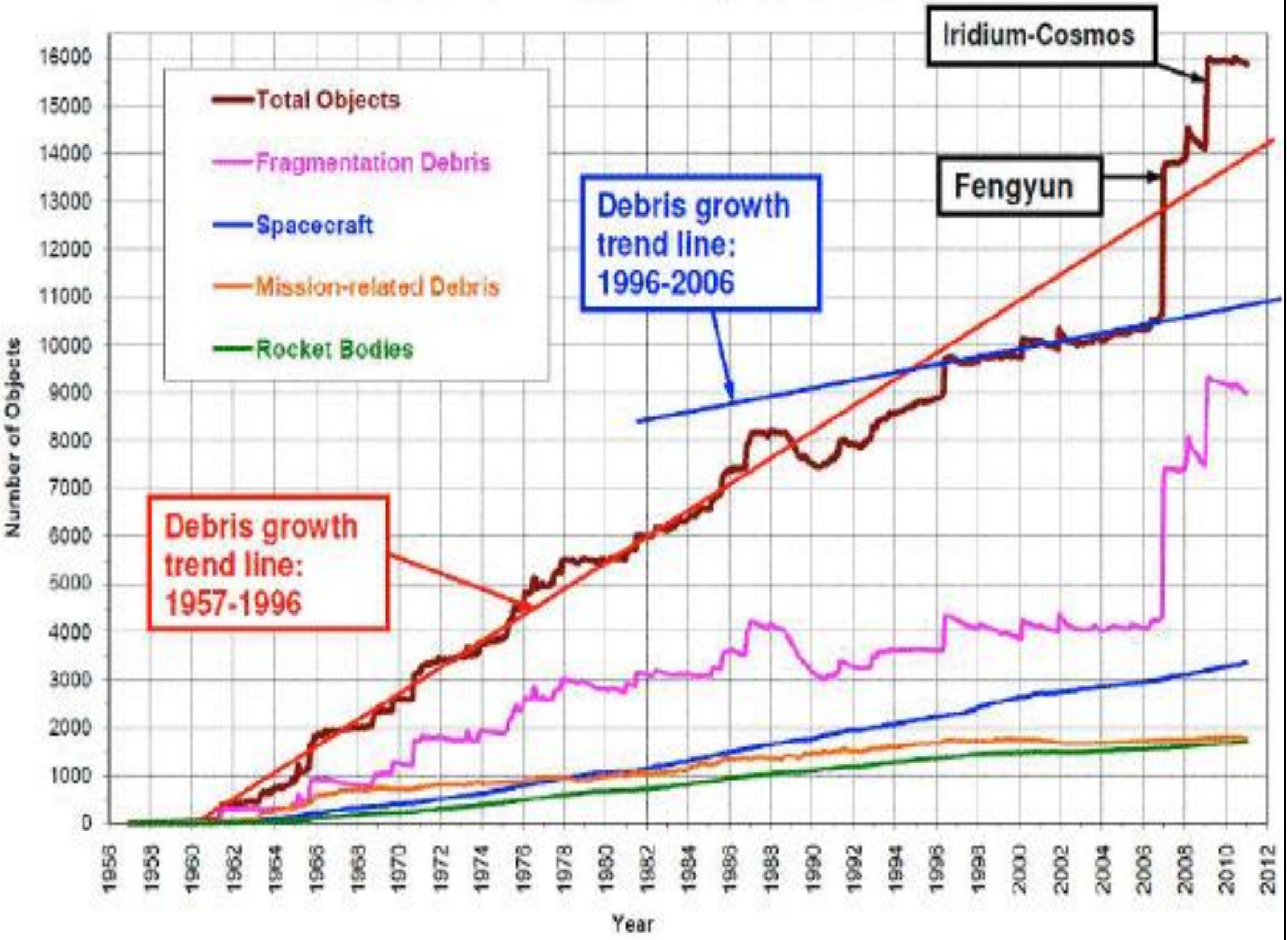









IRIDIUM_4
IRIDIUM_13
IRIDIUM_58
IRIDIUM_8
IRIDIUM_43
IRIDIUM_66
IRIDIUM_50
IRIDIUM_57
IRIDIUM_7
IRIDIUM_18
IRIDIUM_65
IRIDIUM_56
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IRIDIUM_42
IRIDIUM_32
IRIDIUM_46
IRIDIUM_45
IRIDIUM_63

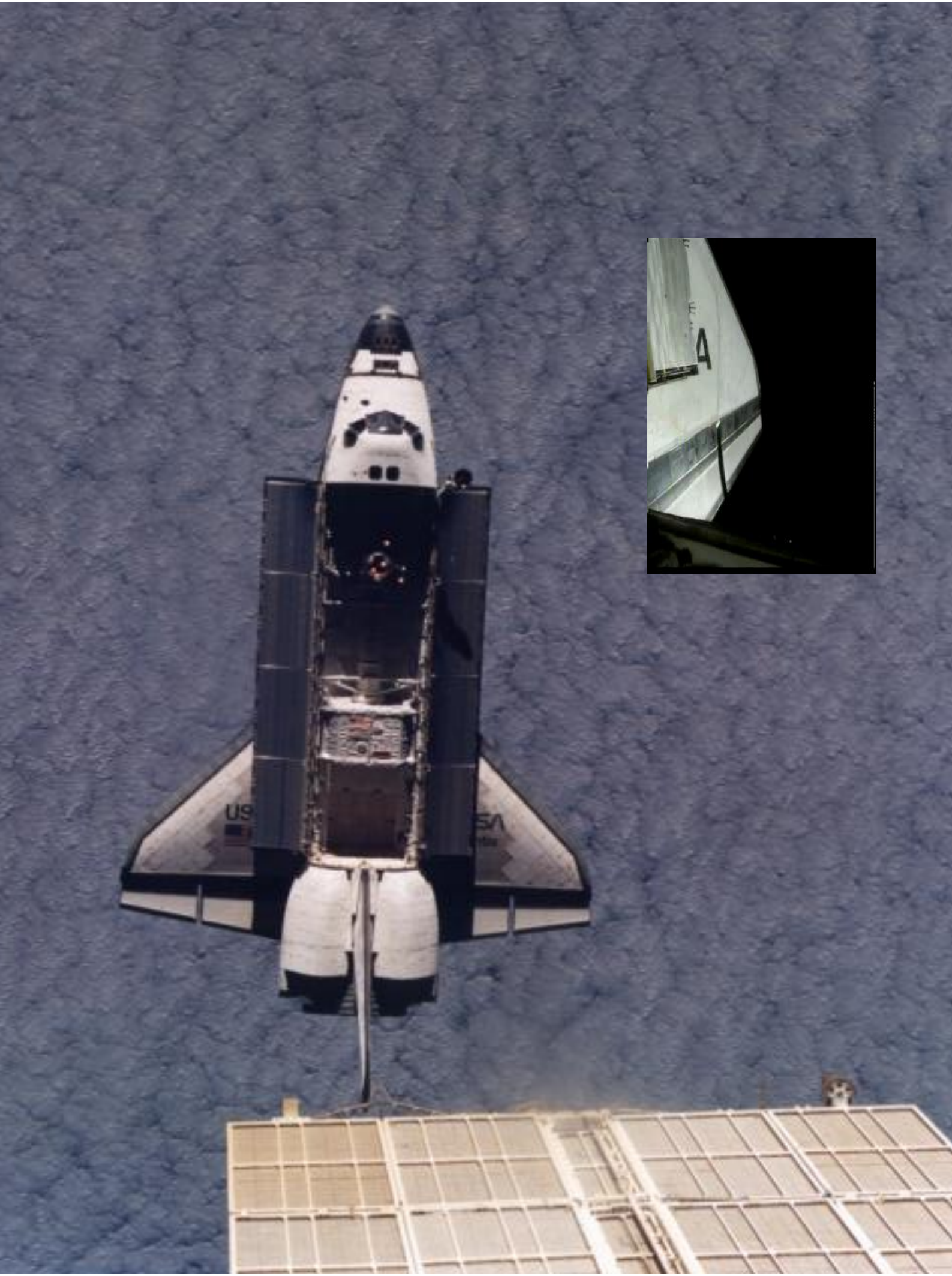

Iridium-Cosmos
Fengyun
Total Objects
Fragmentation Debris
Spacecraft
Mission-related Debris
Rocket Bodies
Debris growth trend line: 1996-2006
Debris growth trend line: 1957-1996
Number of Objects
Year
0
1000
2000
3000
4000
5000
6000
7000
8000
9000
10000
11000
12000
13000
14000
15000
16000
1956
1958
1960
1962
1964
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2012

CASE STUDY 1:
SPACE SHUTTLE ORBITER

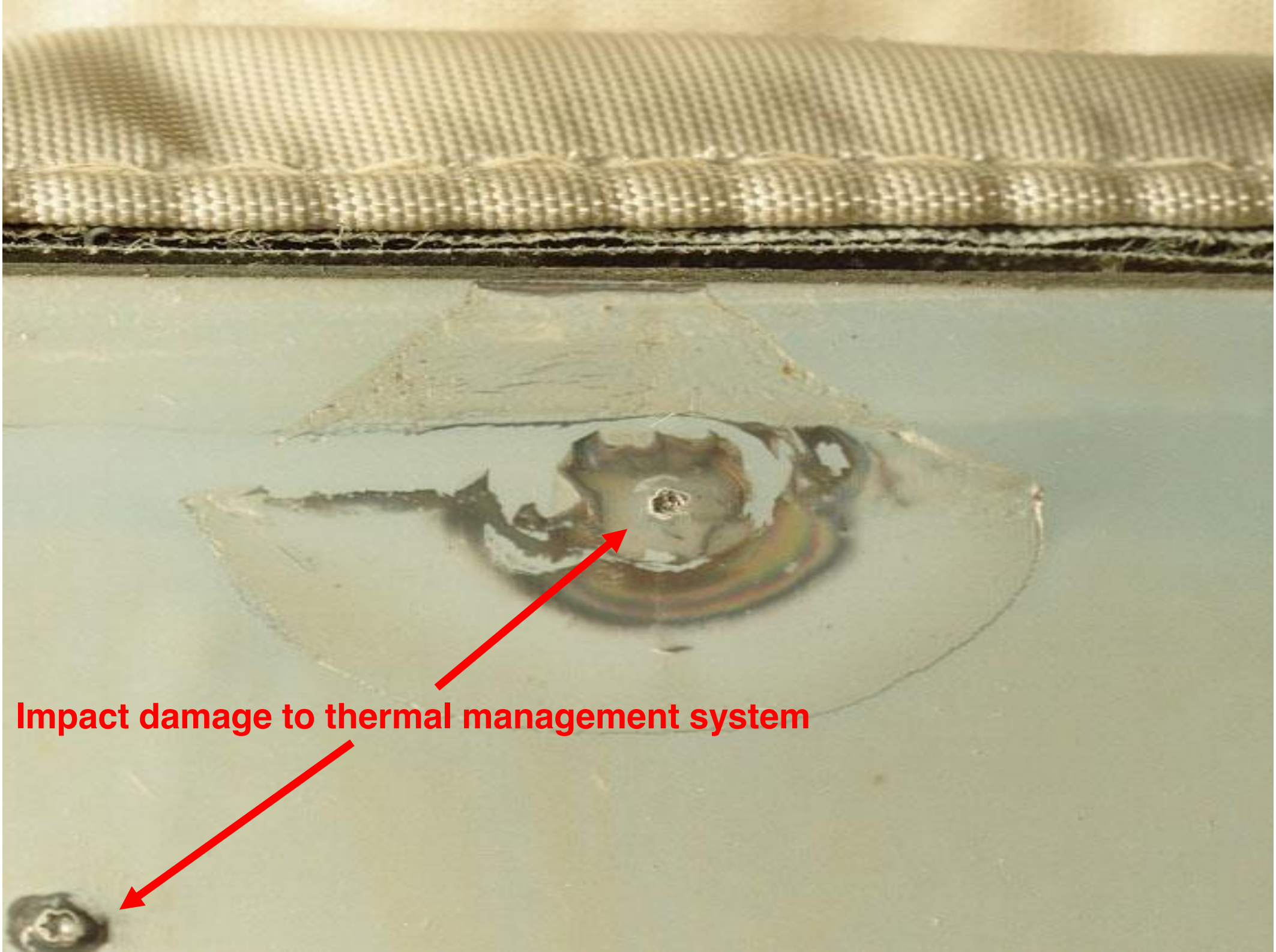
Protecting the Shuttle

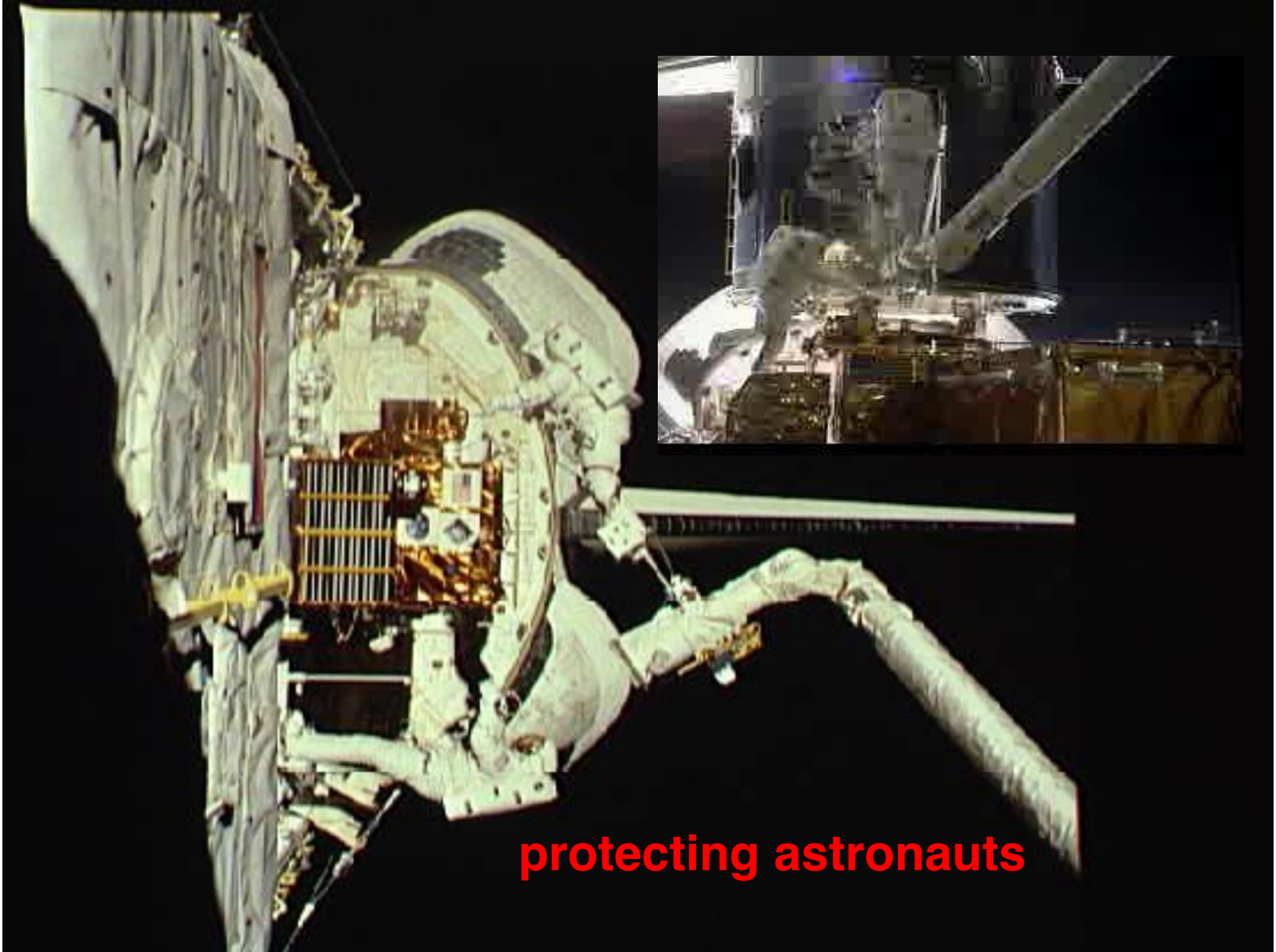


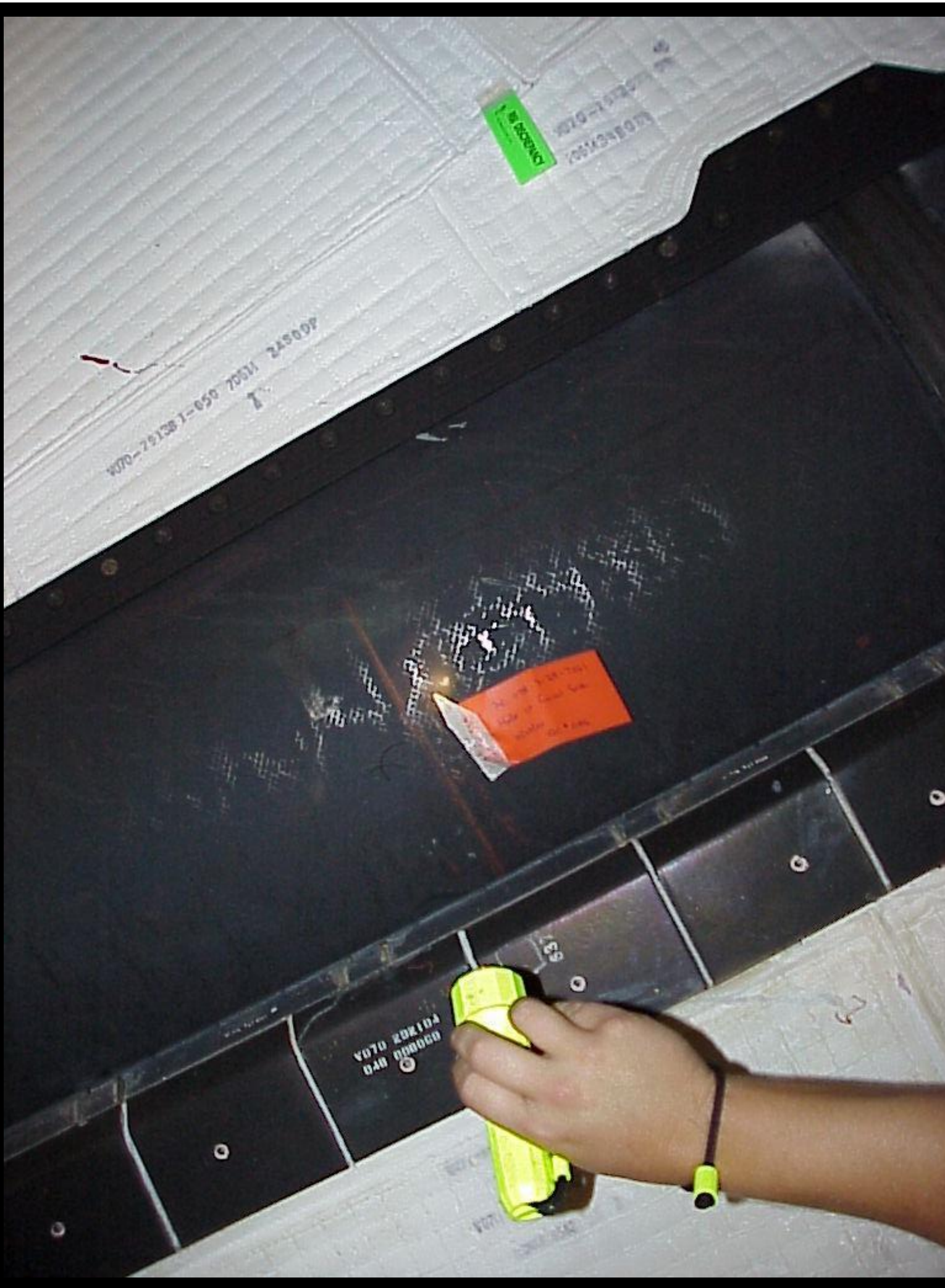







Impact damage to thermal management system

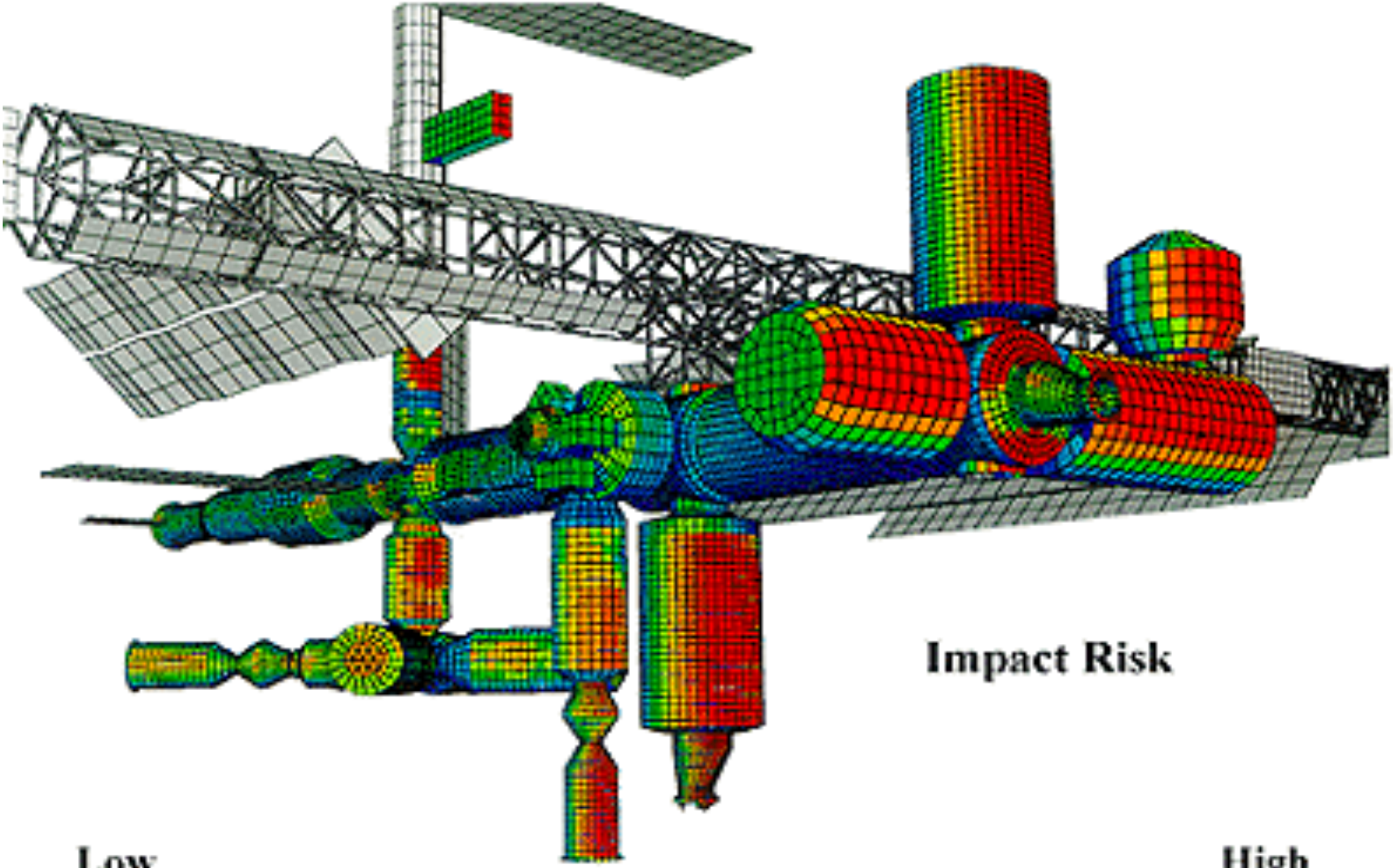

protecting astronauts



CASE STUDY 2: INTERNATIONAL SPACE STATION

International Space Station

Probability of No Impacts From a > 1 cm Ø Debris

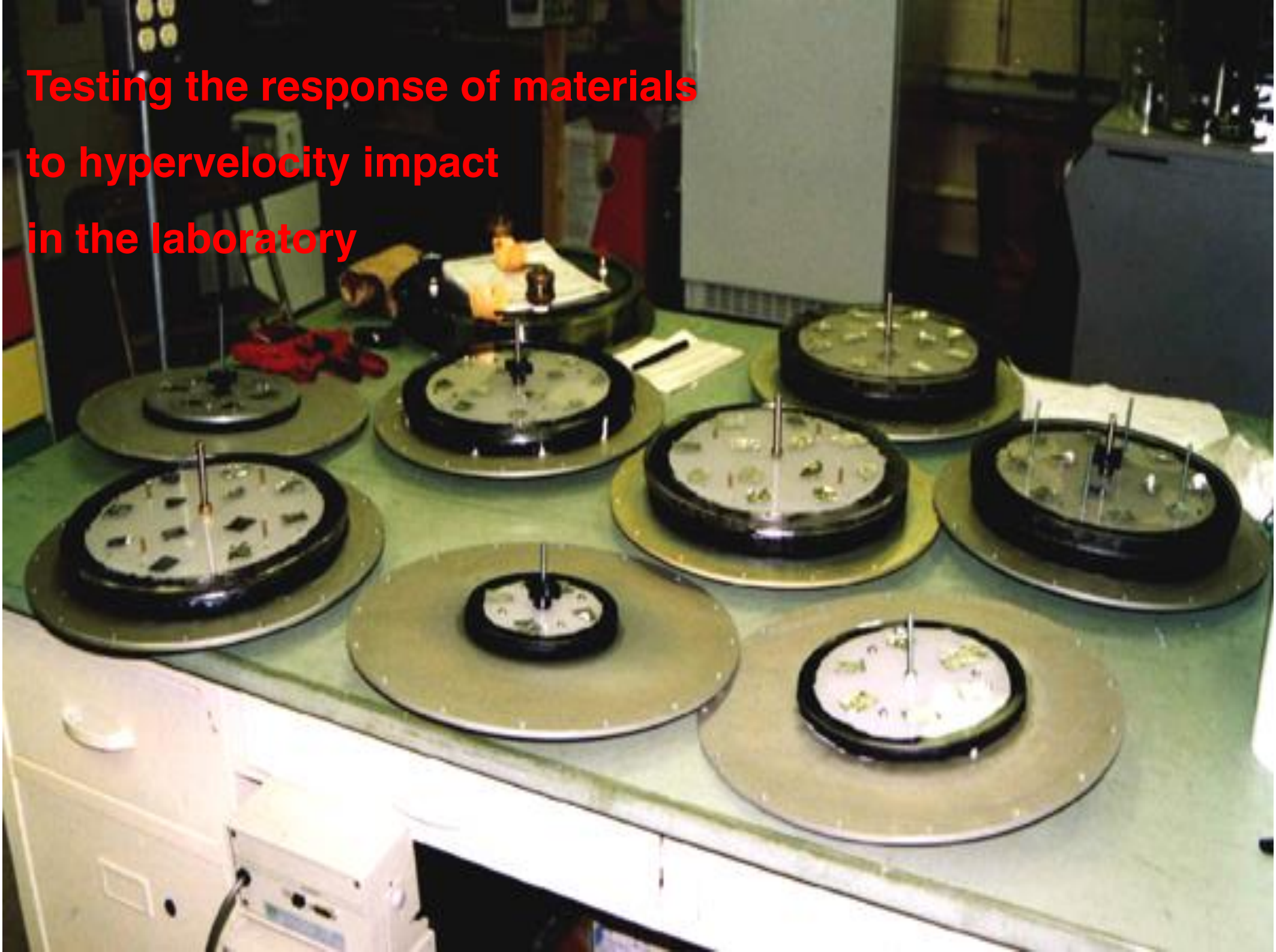


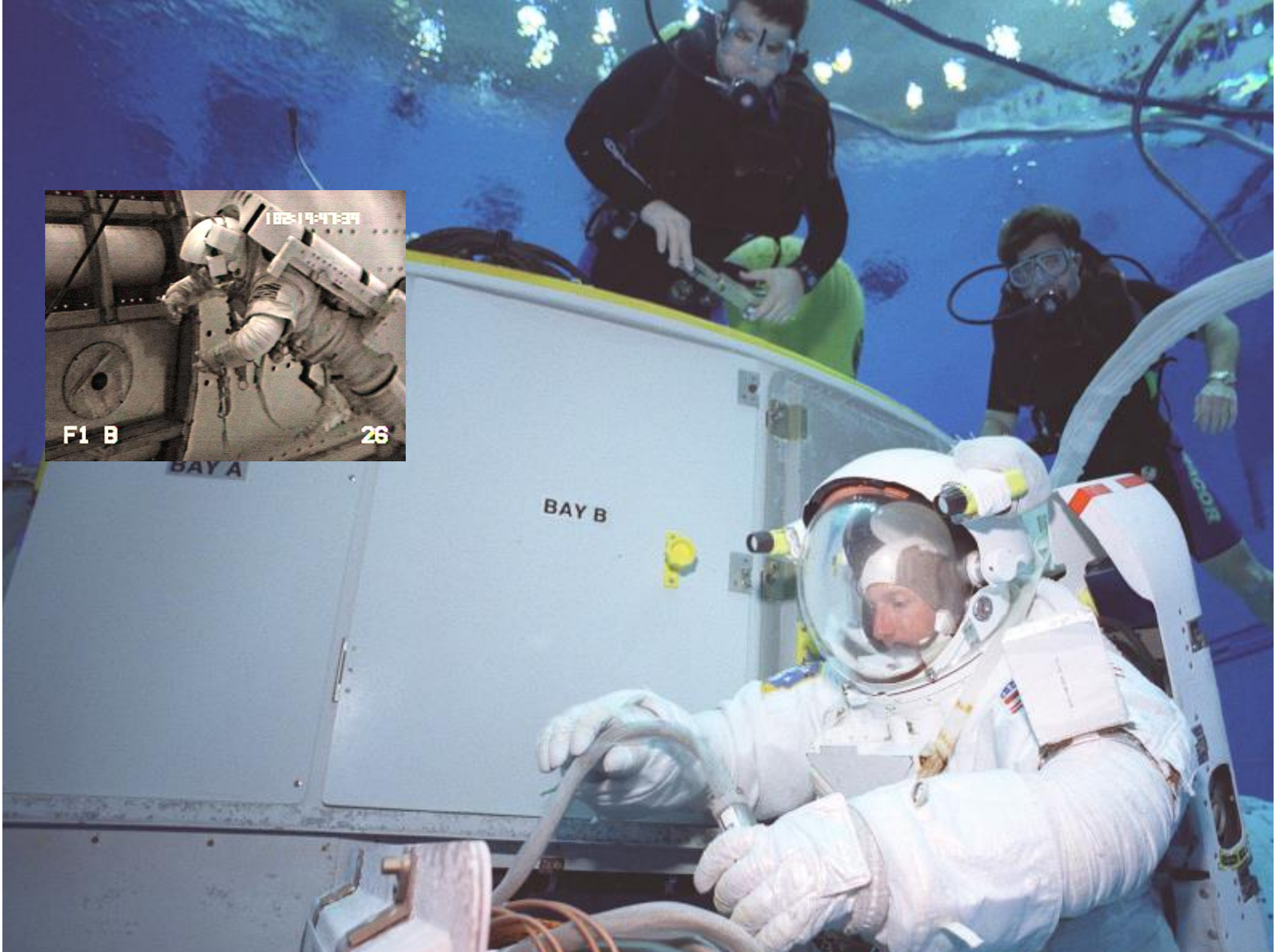


Increase thickness of skin to prevent unzipping

Ensure that complex does not spin up at >2^0/sec

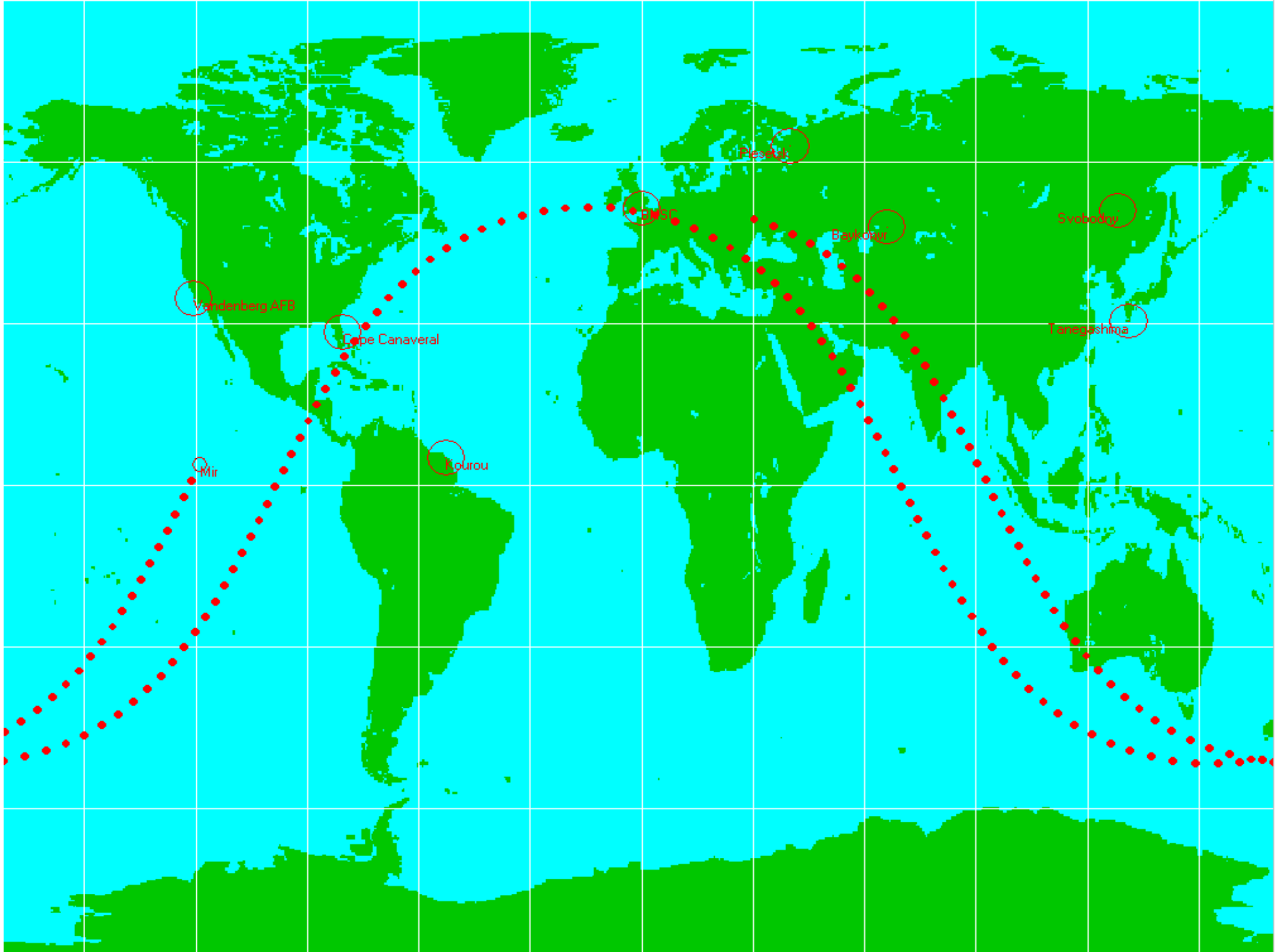
Assess crew survivability following impact


Testing the response of materials
to hypervelocity impact
in the laboratory

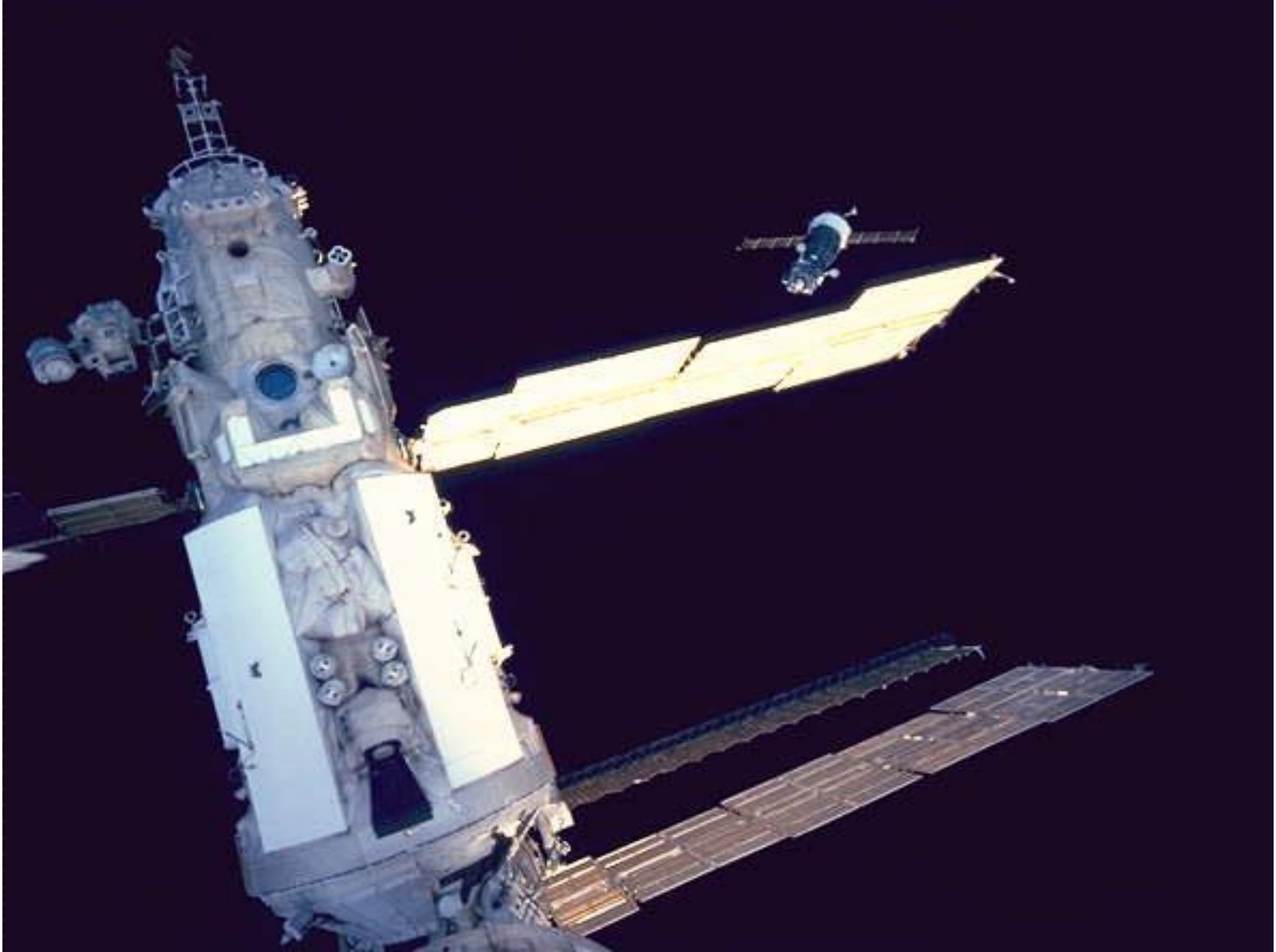

182:19:47:39
F1 B
26
BAY A
BAY B

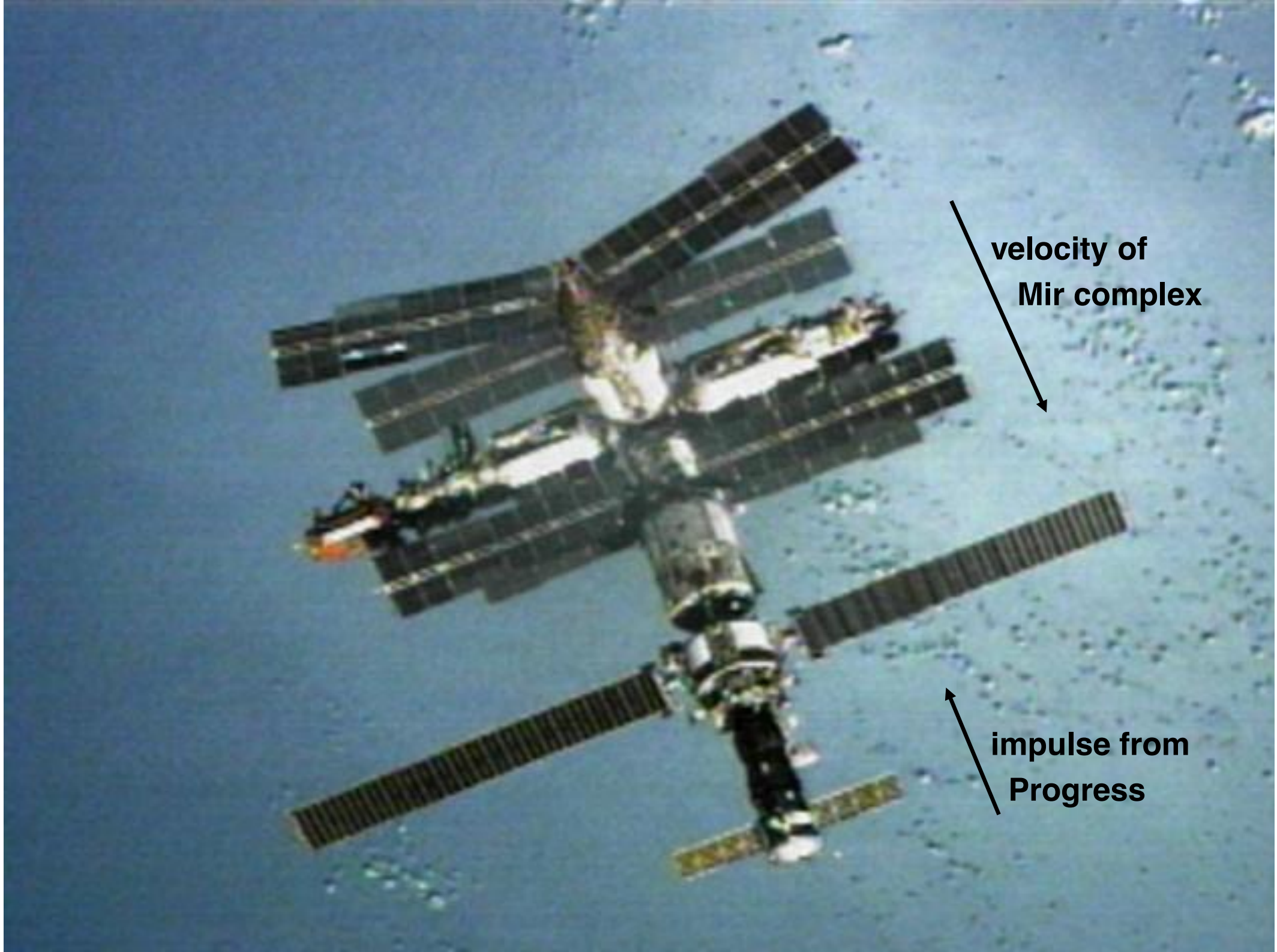
CASE STUDY 3:
MIR SPACE STATION



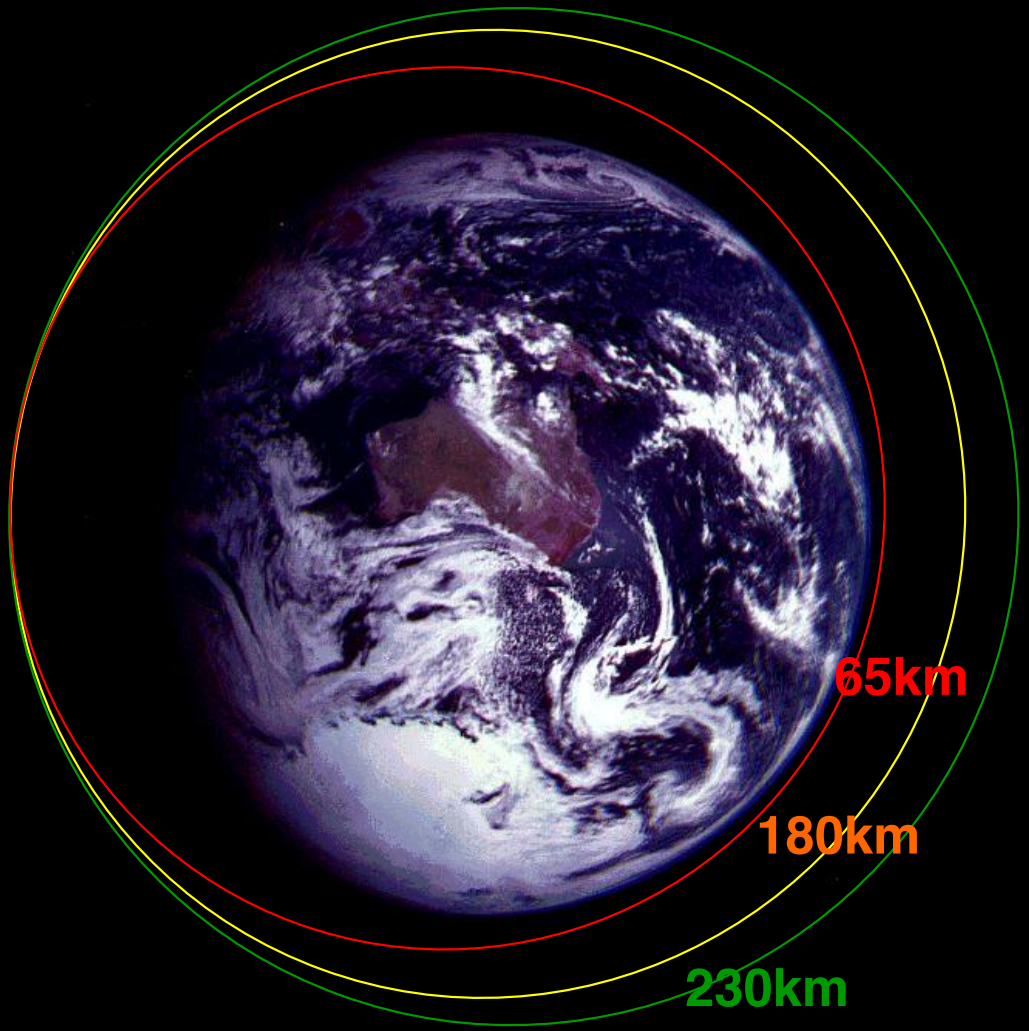

Plesetsk
Baykonur
Svobodny
Vandenberg AFB
Cape Canaveral
Tanegashima
Kourou
Mir


January 24 2001: *Progress* launched from Baikonur



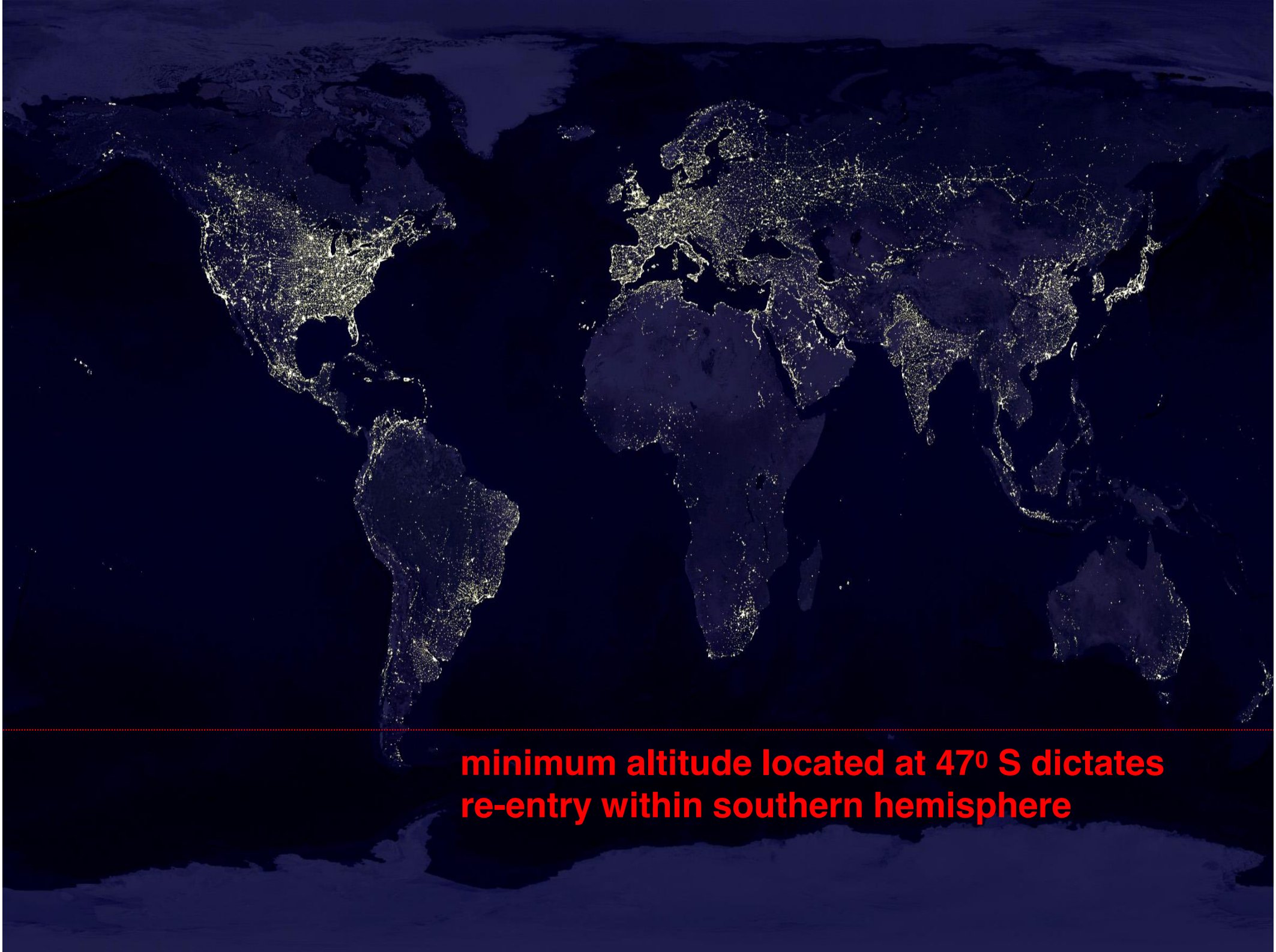

velocity of
Mir complex
impulse from
Progress

Series of manoeuvres to reduce altitude



phasing of manoeuvre around orbit to determine latitude

not to scale

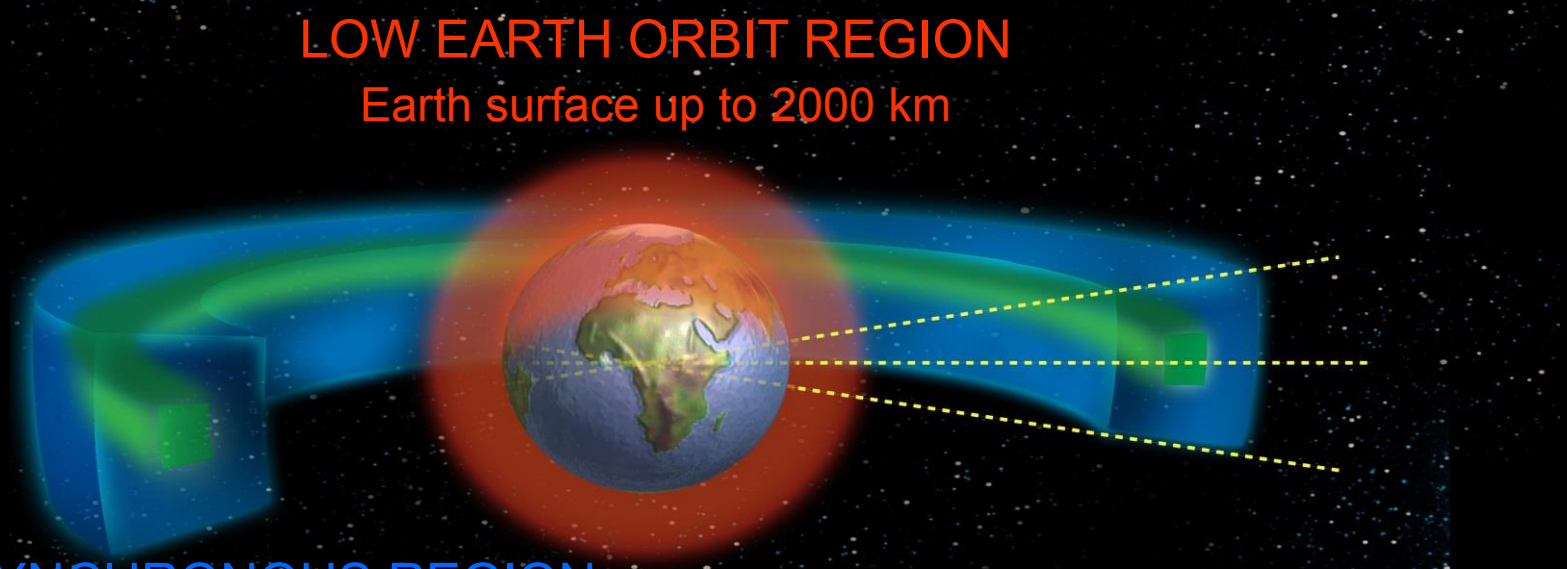

minimum altitude located at 47⁰ S dictates
re-entry within southern hemisphere

Debris Mitigation

- Managing the debris environment does work
- Requires information of orbital population
- Need to share experience between operators
- Effectiveness of measures can be demonstrated
- Need comprehensive implementation to be effective
- Increasing reflected in national legislation
- Regulators need to assess compliance

Definition of Protected Regions

- **Activities in space should recognise the unique nature of 2 regions in space:**



cnes
CENTRE NATIONAL D'ETUDES SPATIALES


AP





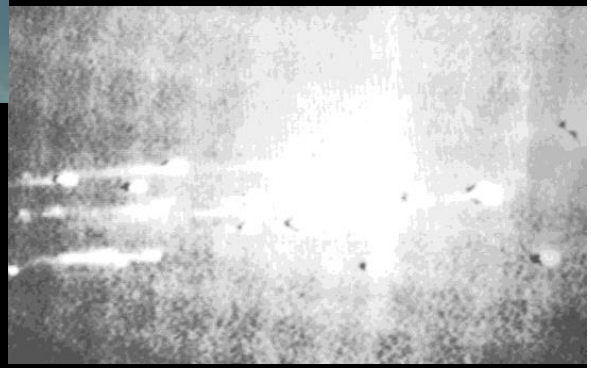





















FRIBEL

Lessons Learnt

- Need to focus efforts on “cause” rather than just “effect”
- As environment deteriorates, cost impacts will increase significantly for all users of space
  - Loss of systems
  - Loss of fuel budget and lifetime due to increased manoeuvres
  - Increased demands of space surveillance
- Active management will be necessary
- Best practice needs to become <u>common</u> practice



UK SPACE AGENCY

Space Security

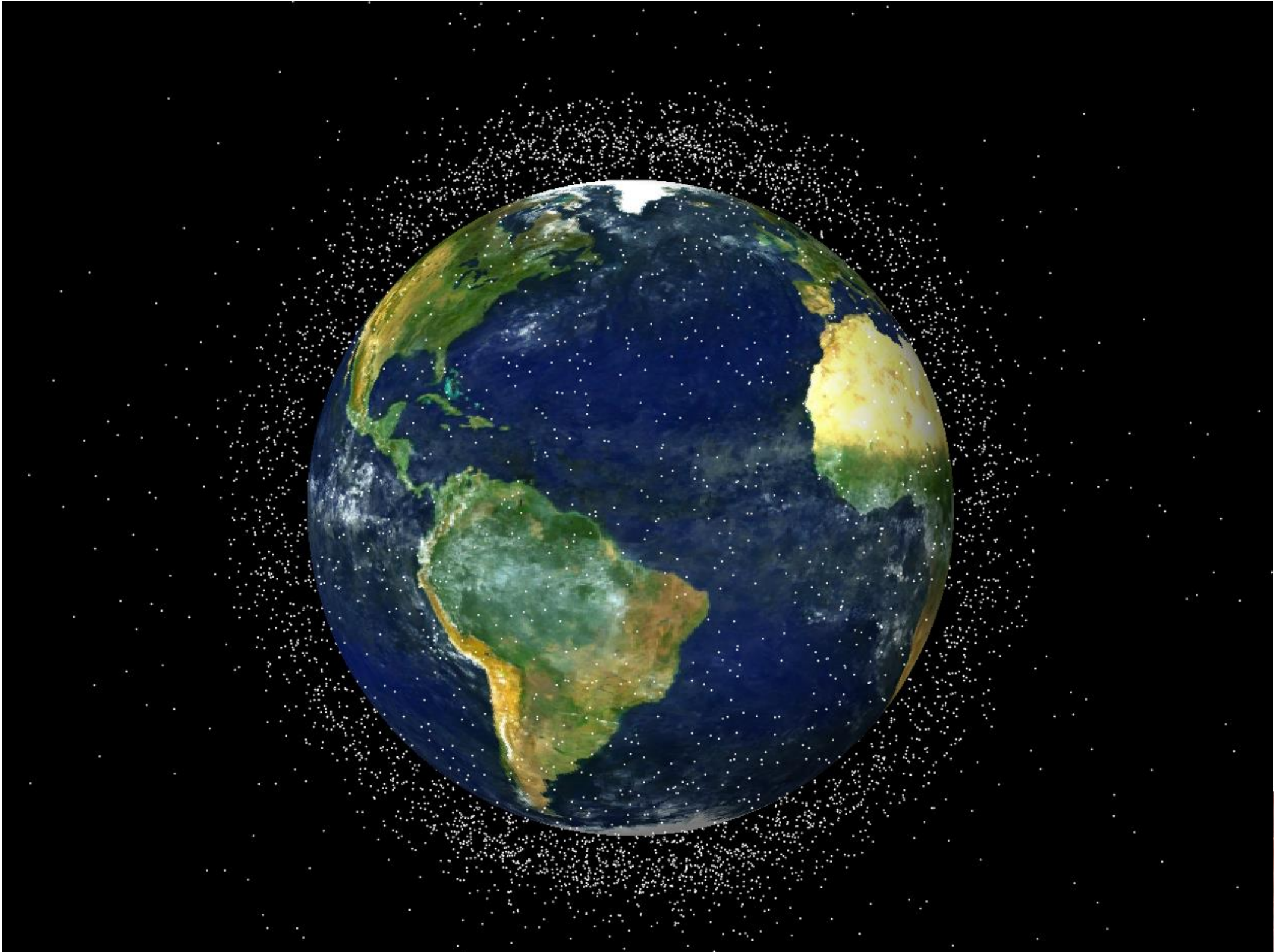
Prof. Richard Crowther
Chief Engineer, UK Space Agency

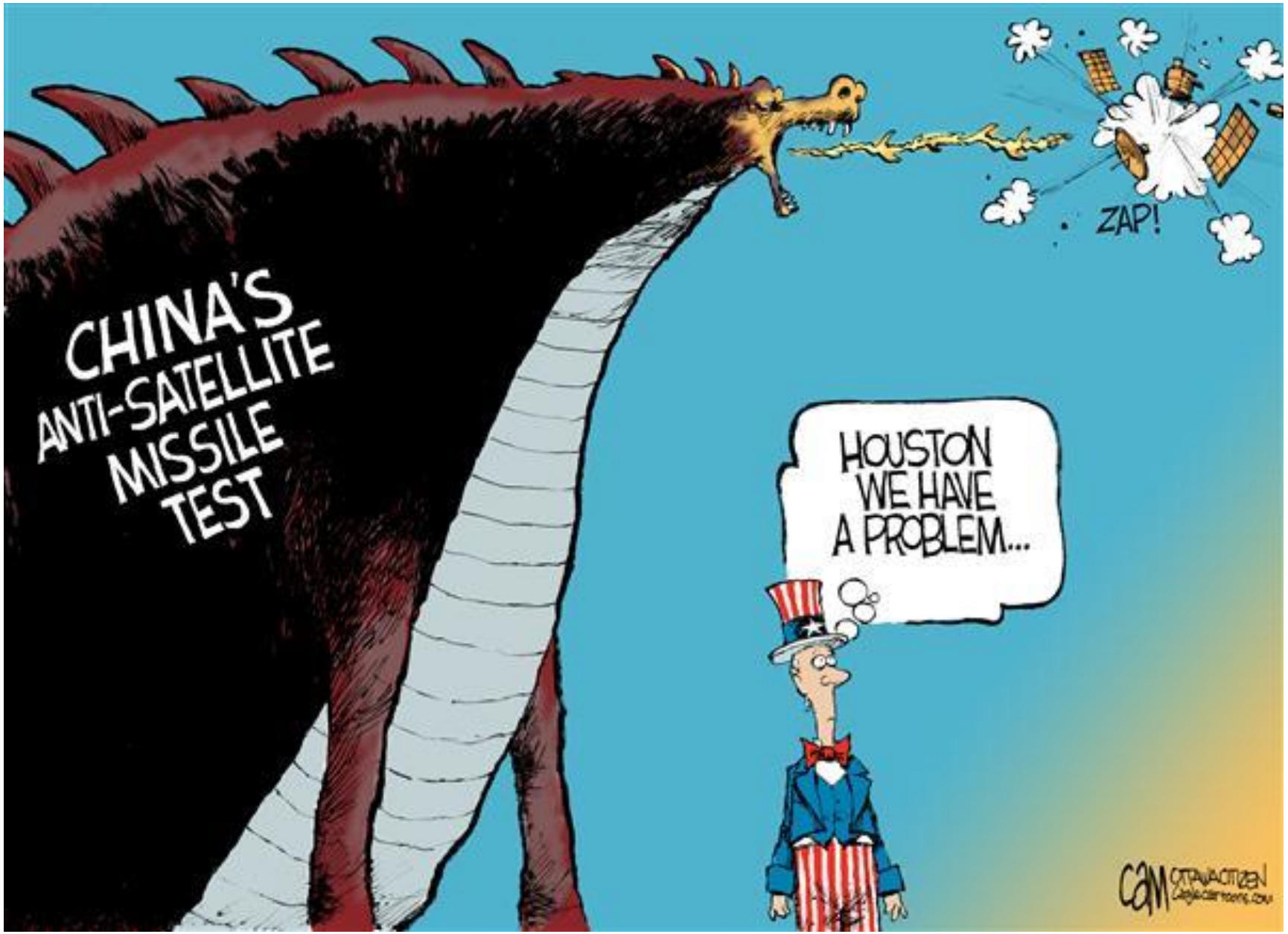
Space is changing

Space is becoming more:

Congested, **C**ontested, **C**ompeted

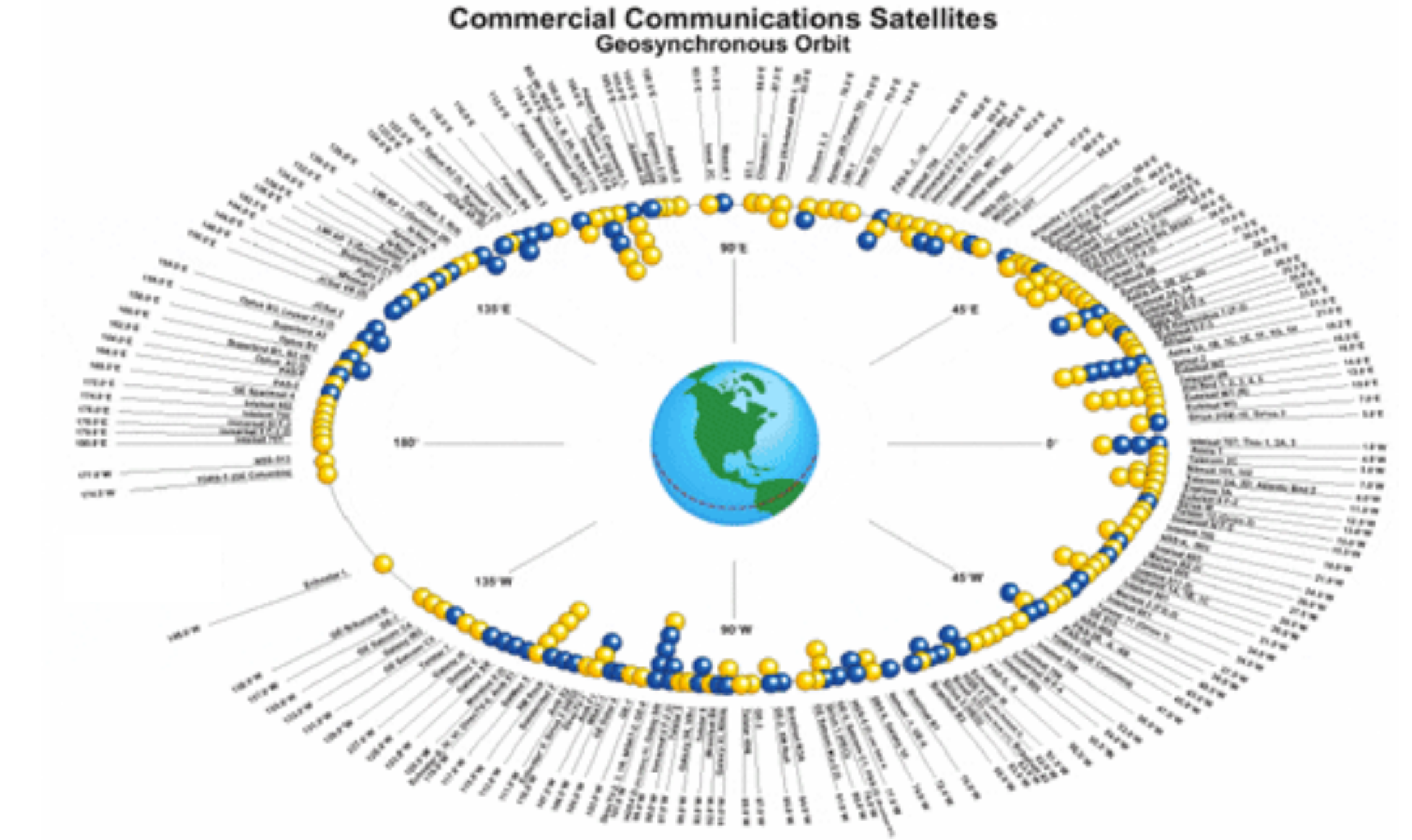





CHINA'S ANTI-SATELLITE MISSILE TEST
ZAP!
HOUSTON WE HAVE A PROBLEM...
CAM OTTAWACITIZEN

Commercial Communications Satellites

Geosynchronous Orbit





Space is changing

Space is becoming more:

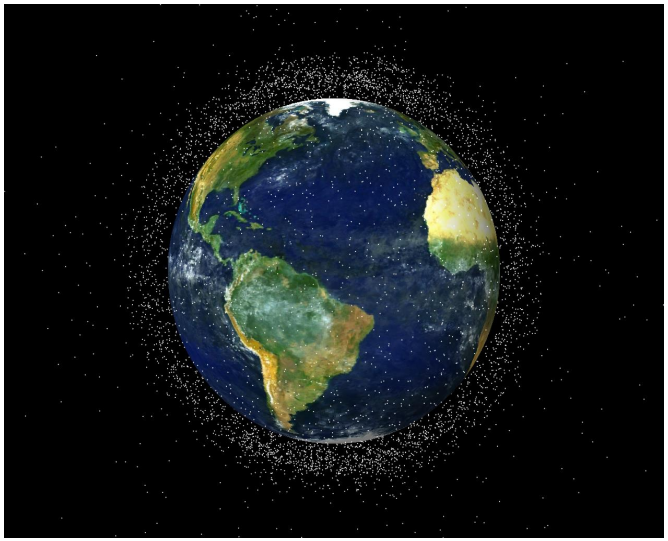
Congested, **C**ontested, **C**ompeted

In response, we need to focus on:

Safety, **S**ecurity, **S**ustainability



Governance Forums



Committee on Peaceful Uses
Of Outer Space (COPUOS)

Conference on Disarmament
(CODUN)

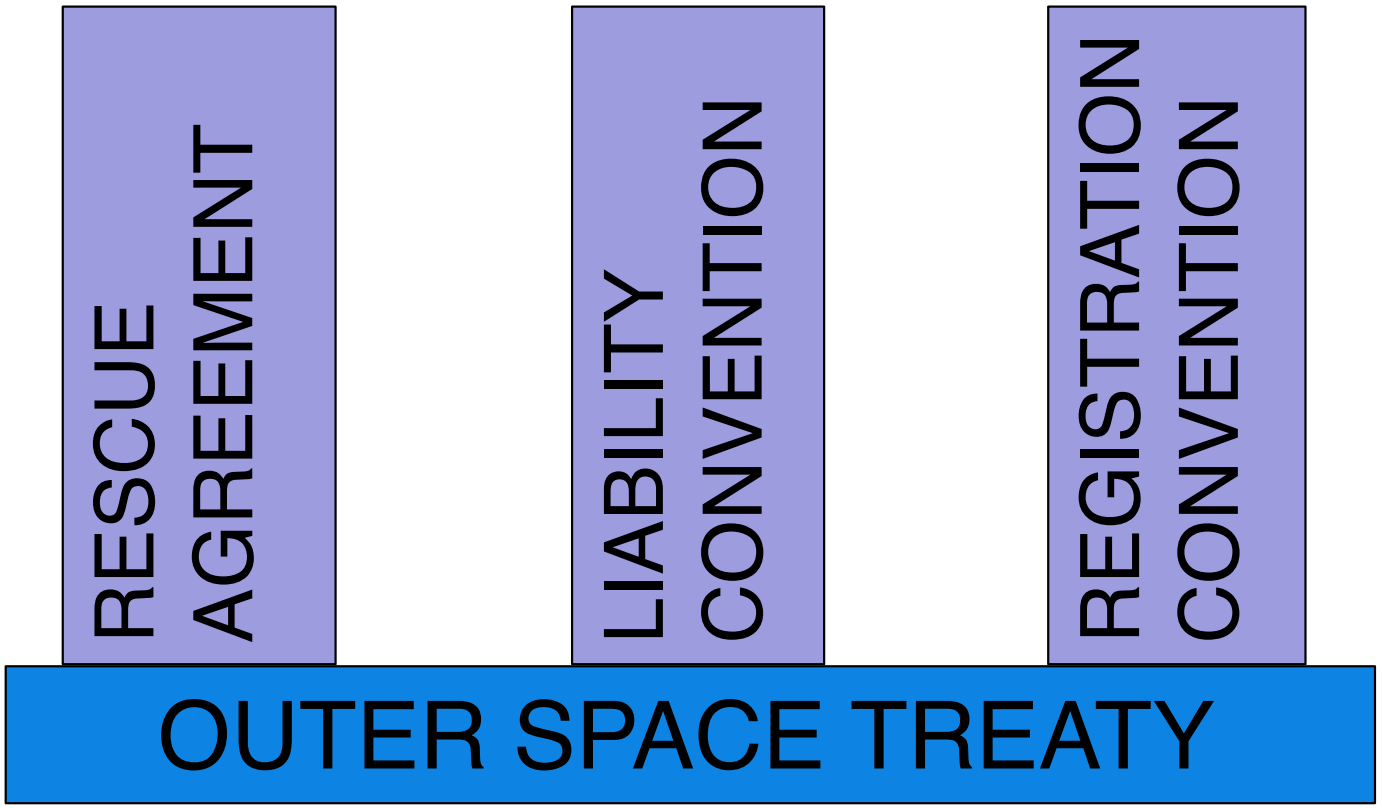


Regulation of Space

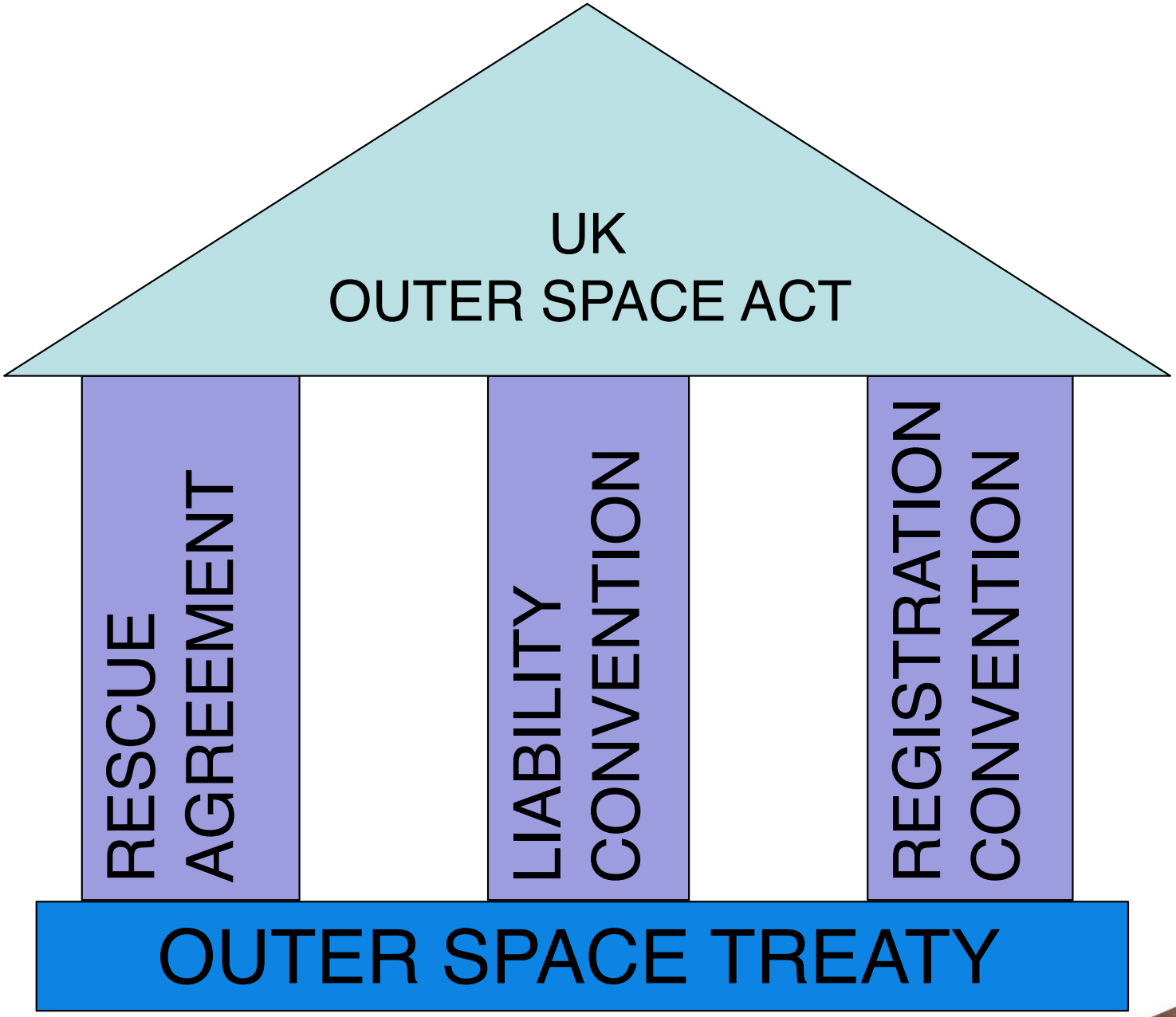
OUTER SPACE TREATY



Regulation of Space



Regulation of Space



OSA LICENSING CONSIDERATIONS

WHO?	WHY?	WHEN?
HOW?	WHAT?	WHERE?



Estimated Orbital Population

Size	Number	% Mass
>10 cm	>17000	99.93
1-10 cm	>400,000	0.035
<1 cm	>35,000,000	0.035
Total	>35,000,000	>6,000 tonnes



Estimated Orbital Population

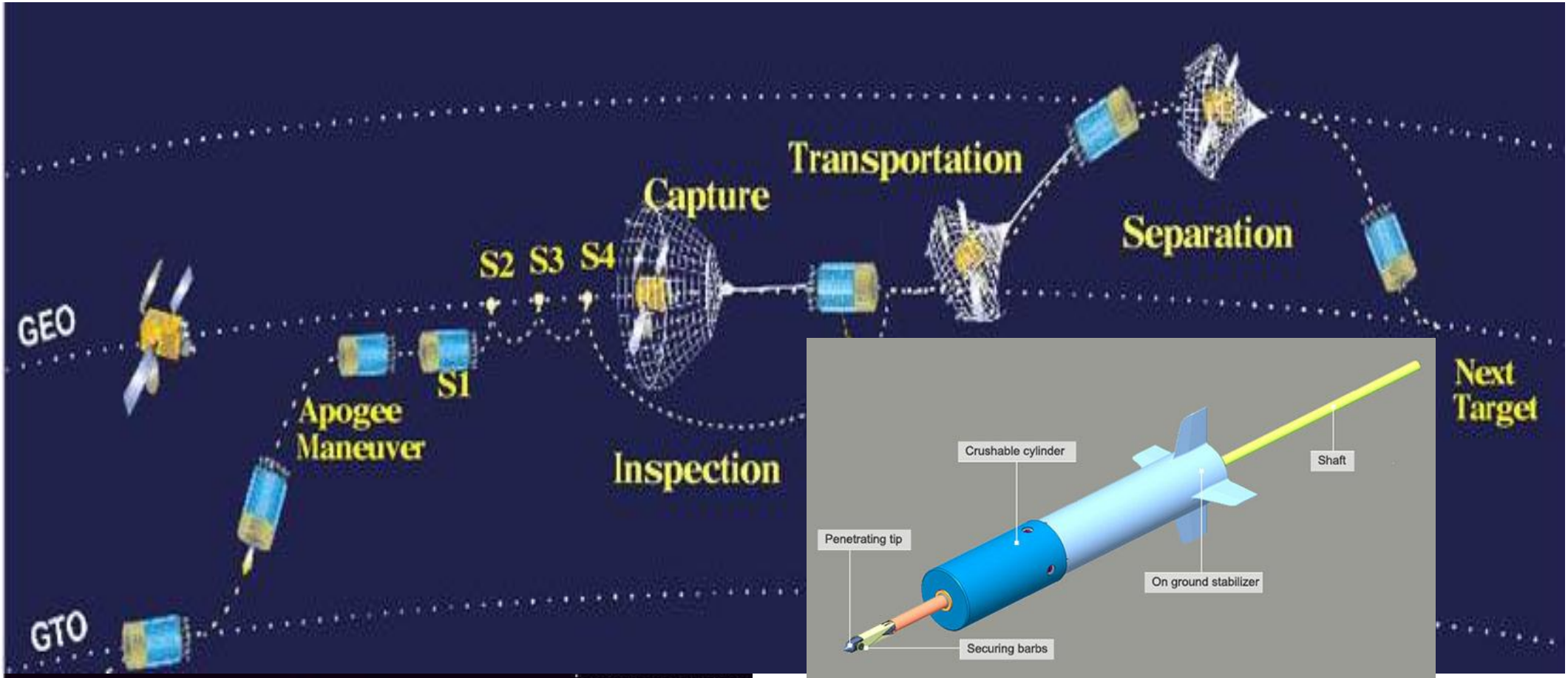
	Size	Number	% Mass	
Tracked objects	**>10 cm**	**>17000**	**99.93**	< 10% operational
	1-10 cm	**>400,000**	**0.035**	
	<1 cm	**>35,000,000**	**0.035**	
	Total	**>35,000,000**	**>6,000 tonnes**	

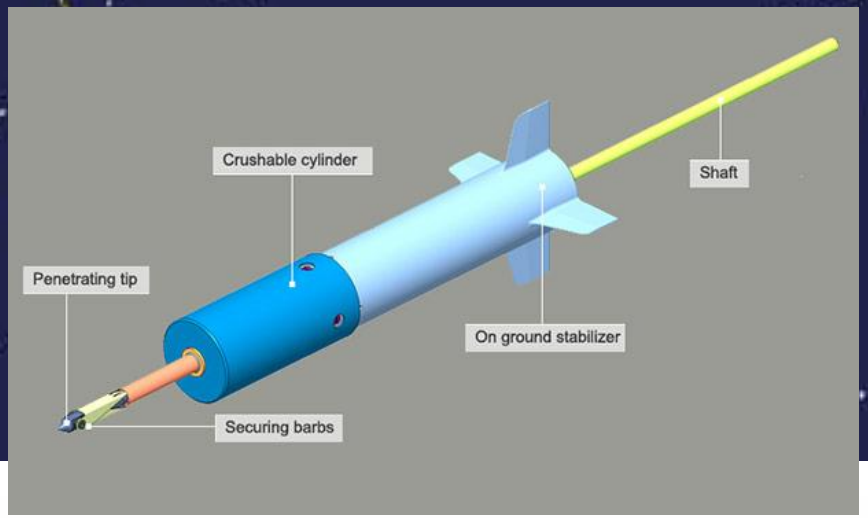


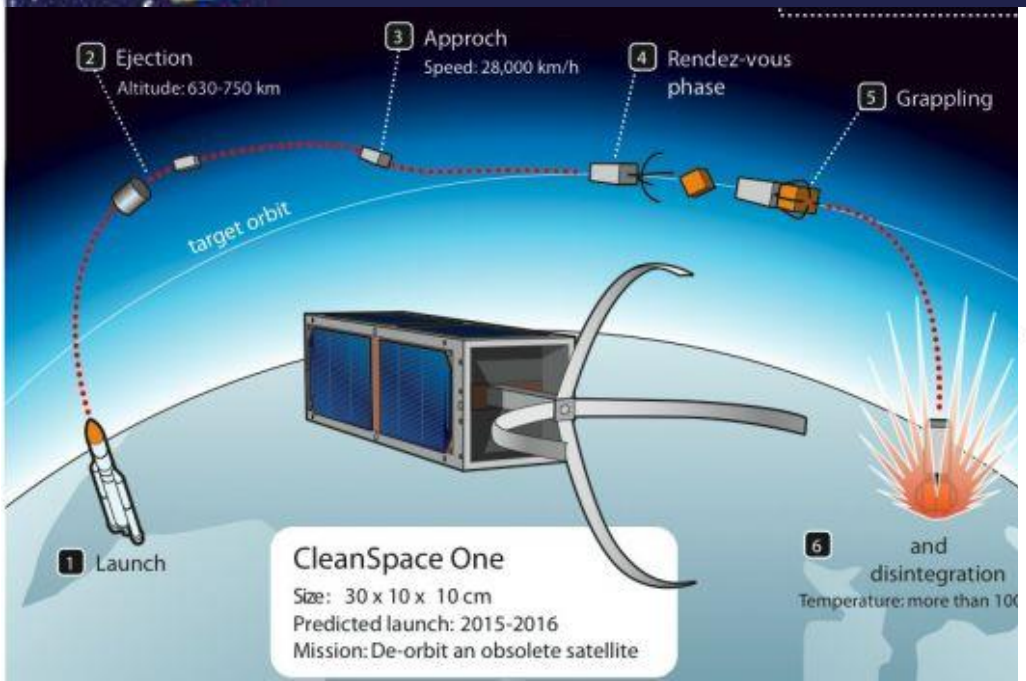
We can only “manage” a small proportion of objects through “mitigation”

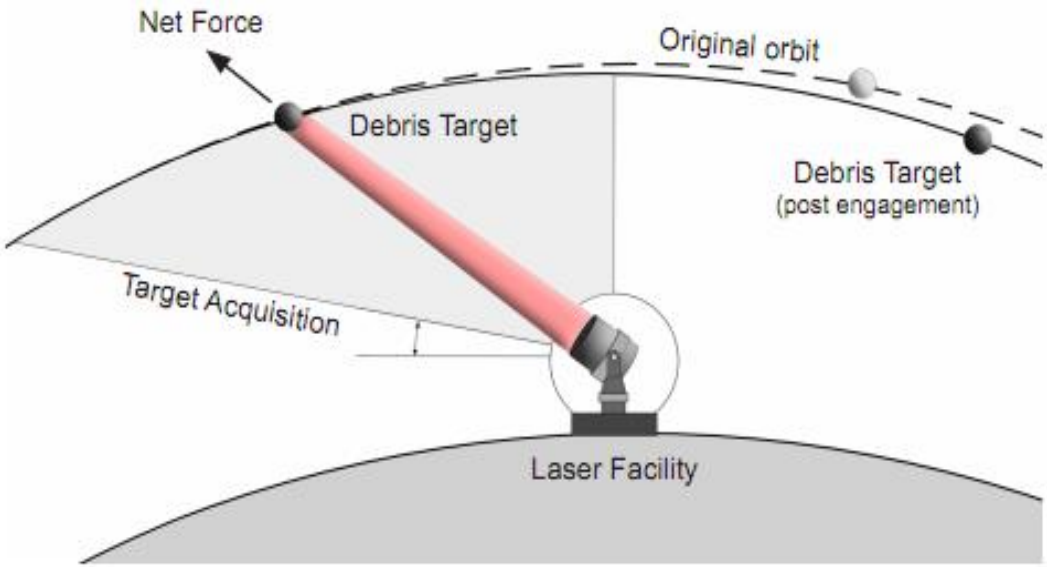
What is the answer?




GEO
GTO
Apogee
Maneuver
S1
S2 S3 S4
Capture
Transportation
Separation
Inspection
Next
Target


Crushable cylinder
Shaft
Penetrating tip
On ground stabilizer
Securing barbs


2 Ejection
Altitude: 630-750 km
3 Approch
Speed: 28,000 km/h
4 Rendez-vous
phase
5 Grappling
target orbit
1 Launch
CleanSpace One
Size: 30 x 10 x 10 cm
Predicted launch: 2015-2016
Mission: De-orbit an obsolete satellite
6 and
disintegration

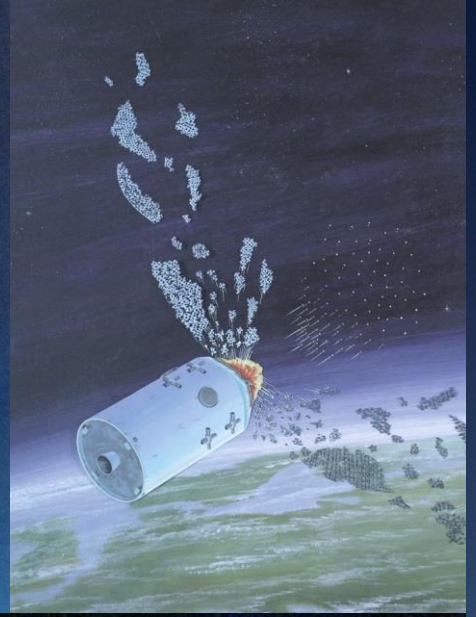

Net Force
Original orbit
Debris Target
Debris Target
(post engagement)
Target Acquisition
Laser Facility

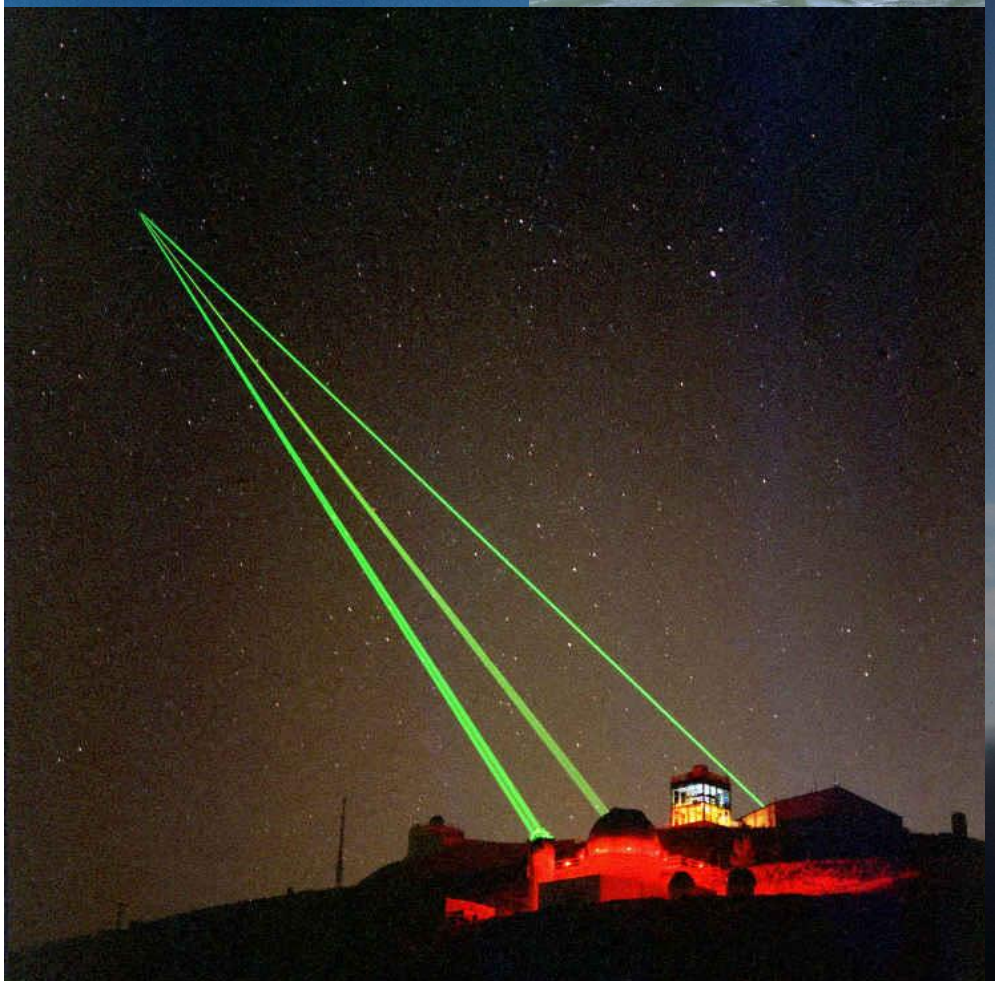
Dual Use Systems

- Dual Use systems can be used for either peaceful or military/security aims, or both
- Rapidly evolving technology has already broadened the scope of potential conflict from land, maritime and air environments to encompass cyber and space domains
- Appropriate consideration needs to be given to the potential of dual use space systems

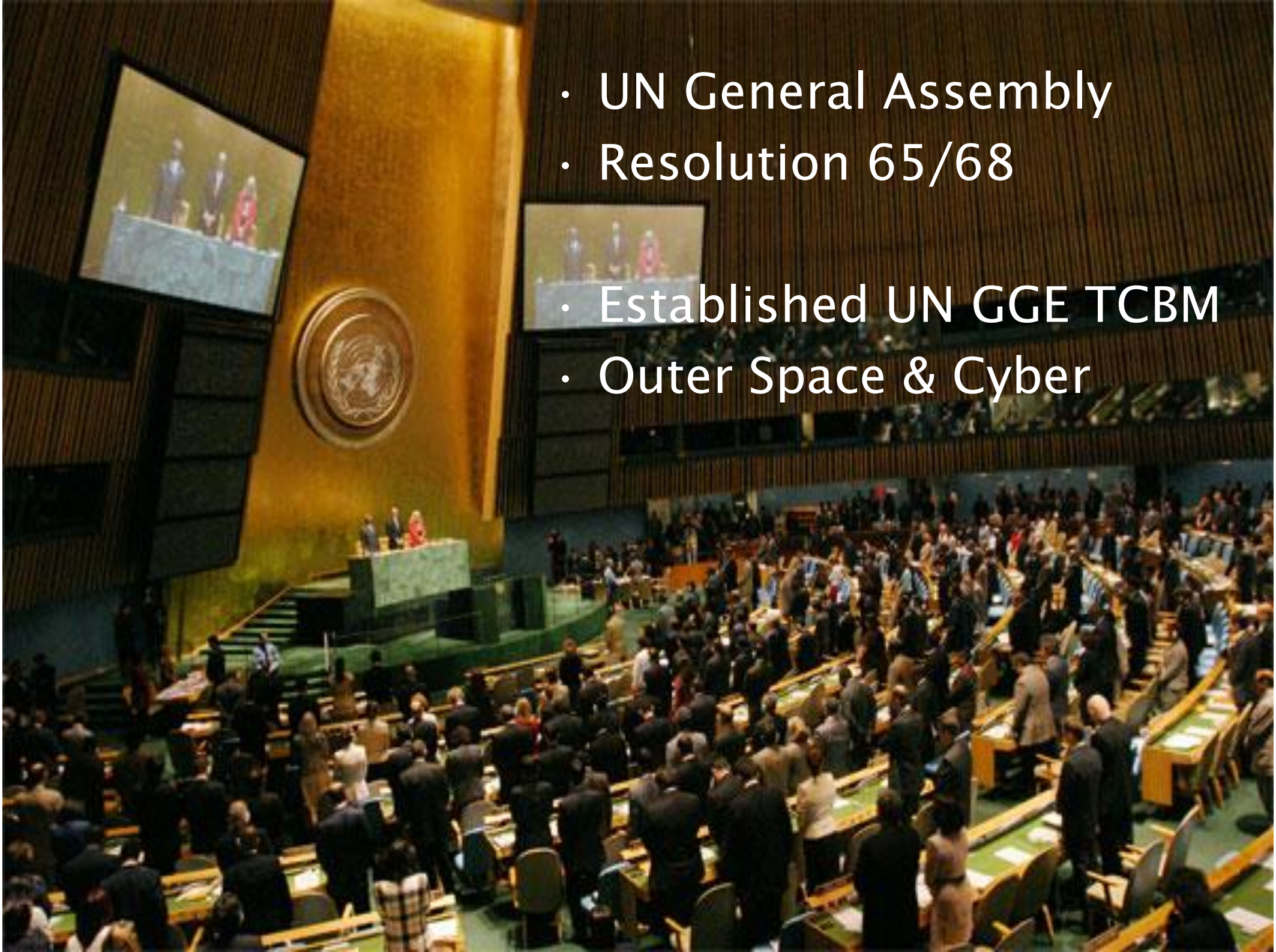












- UN General Assembly
- Resolution 65/68

- Established UN GGE TCBM
- Outer Space & Cyber

UN GGE on TCBMs

UN: United Nations

GGE: Group of Governmental Experts

TCBMs: Transparency and Confidence Building Measures

Aim is to remove potential for suspicion over motives and escalation of hostilities



TCBMs

TCBMs include:

- Treaties (Anti-ABM)
- Forums (IADC)
- Codes of Conduct (Hague/EU?)
- Removal of capability (SORT)
- Inspections
- Workshops
- Exchanges





The Group of Governmental Experts on Transparency and Confidence-Building Measures in Outer Space Activities, United Nations, New York, 27 July 2012

From left to right: João Marcelo Galvão de Queiroz - Brazil, Chulmin Park - Republic of Korea, Richard Crowther - United Kingdom, Musthafa M. Jaffeer - Sri Lanka, Peter Martinez - South Africa, Andrii Kasianov - Ukraine, Frank Rose - USA, Victor L. Vasiliev - Russian Federation - Chairman, Augustine U. Nwosa - Nigeria, Gérard Brachet - France, Dumitru-Dorin Prunariu - Romania, Sergio Marchisio - Italy, Huaicheng Dai - People's Republic of China, Hellmut Lagos - Chile, Ruslan Amirgereyev - Republic of Kazakhstan

Tiered approach to security of space

- Aim is to deter potentially damaging activities in outer space through:
  - Establishment of norms of behaviour to identify irresponsible actions
  - Building of international partnerships to forge alliances and encourage inter-dependency
  - Development of resilient systems to mitigate consequences of impacts on systems
  - Enabling self-defence to respond to hostile actions/actors



Summary

Our goal is a stable space environment.

However, space is becoming more:

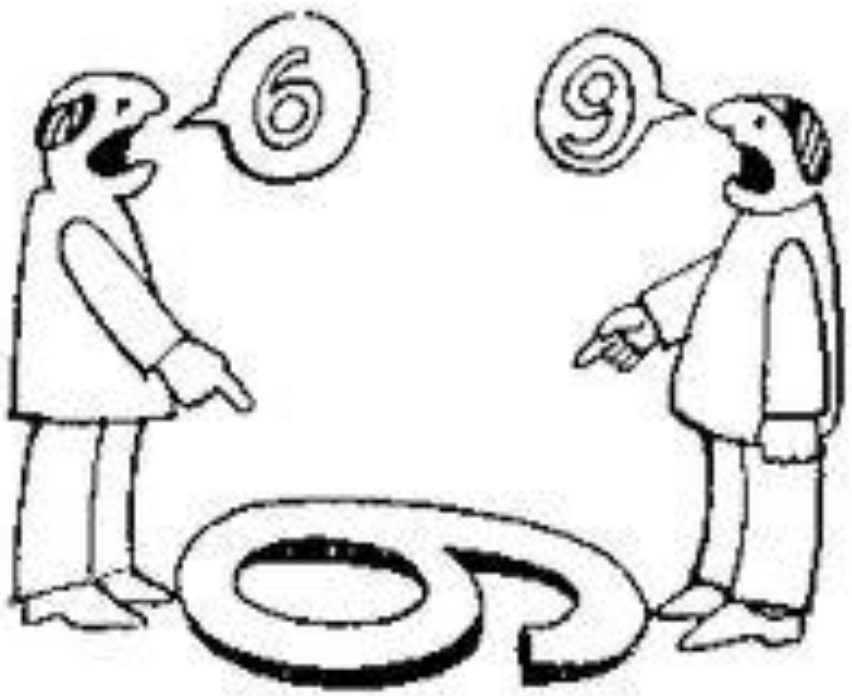
Congested, **C**ontested, **C**ompeted

In response, we need to focus on:

Safety, **S**ecurity, **S**ustainability



Summary



Safety + **S**ecurity=**S**ustainability

